

Applications of evolutionary game theory in urban road transport network: A state of the art review

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ABSTRACT

A sustainable transport infrastructure is one of the pillars of a sustainable city. However, the literature indicates that urbanization, population growth, changes in population density, and motorization make it difficult for the current road transport system to meet mobility needs for a sustainable city. Traffic crashes and congestion on roads are common as a result of increasing travel times, fuel consumption, and carbon emissions, thereby reducing efficiency and sustainability of mobility systems. Managing these issues involves the interaction of multiple decision-makers, such as vehicles, pedestrians, traffic system operators, and authorities. Accordingly, these are well-suited to being analyzed under the guise of game theory. While classical game theory possesses multiple limitations, it can be argued that evolutionary game theory (EGT) models are more effective for real-world scenarios. This manuscript presents a state-of-the-art review on EGT applied to the road transportation network. The manuscript has divided the application of EGT in advancing the transportation network into multiple categories, i.e., choice-based analysis, traffic management, behavioral interactions, routing operation, and transport safety. This manuscript provides an in-depth analysis and a comparative criticism of the various proposed evolutionary game models. Finally, the manuscript discusses the challenges and provides recommendations for future research on evolutionary game models in transportation networks. These insights aim to facilitate targeted activities based on current research needs.

1. Introduction

Transport networks play an essential role in the efficient operation of megacities. However, the present-day traffic network operates inefficiently, causing several adverse environmental effects and deteriorating the quality of life for users (Elvik, 2014). Budget and ecological restrictions sometimes make building new roads or improving existing infrastructure challenging. Thus, it has become a crucial research area to develop efficient traffic and transportation management and control systems that can satisfy an ever-increasing demand for mobility (Zhu et al., 2022).

Transportation research is centered on a complex real-world system. In addition to providing a detailed overview of the theoretical principles and practical techniques involved in planning, designing, operating, and transport safety, other aspects have also been studied. A distinctive characteristic of transport research is that it evolves extensively with technological innovation. For a long time, professionals and scholars

have been interested in developing a better understanding of transport network systems, and their temporal changes. Ever since Euler (1736) (Shields, 2012) introduced the initial transportation network in the famous problem of the seven bridges of Königsberg, extensive advancements have been made in transportation research, encompassing a wide range of areas within the transportation system. These advancements have greatly expanded the scope and depth of our understanding.

In transportation engineering, major research topics include travel behavior, traffic flow, transport networks, traffic control and management, freight systems, and other modes of transportation (Yanf, 2005). Traffic flow is studied from a micro-to-meso-to-macroscopic perspective and includes human factors, autonomous vehicles, and many others. Studies based on empirical evidence and modeling based on statistical and computer science principles are among the most common approaches (Antonioni et al., 2015). Travel behavior studies include the analysis of demand, route choice, day-to-day dynamics, mode choice, lane selection, and activity choices (Adler et al., 2021). Managing and

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controlling traffic involves designing and managing traffic control devices, congestion pricing, traffic routing, route guidance systems, traveler information packages, and, more recently, vehicular communication systems. There are usually optimization and control methods involved in this process (Sakoguchi et al., 2019). Traffic flow and traveler behavior at signalized and non-signalized intersections, as well as driver behavior, constitute the transportation network. Planning and operations models are usually integrated with design and performance evaluation (Guo et al., 2019). Additionally, research objects include safety in transportation, and several alternative modes of transportation, including public transit, bicycles, and pedestrians, which are both crucial components of transportation systems and can be studied in conjunction with behavior models, operational models, or passenger vehicles as alternatives (Hassan et al., 2014). Throughout the history of transportation research, these basic research objects have remained relatively static; however, models for describing and analyzing them have evolved.

In some sense, behavioral interactions between humans and technologies constitute the basis of the entire transportation system. Research effort needs to be carried out on building algorithms capable of recognizing human decisions and interactions to solve these problems. In the past, planning and implementation of behavioral and decision-making strategies have been solved using classical game theory. Although very few researchers have worked on a state-of-the-art review about GT's role in advancing transportation networks. The comparative literature of GT from various aspect of transportation network including travel choices (Elvik, 2014; Zhu et al., 2022; Adler et al., 2021; Di & Liu, 2016; Ji & Levinson, 2020; Hollander & Prashker, 2006; Klein & Ben-Elia, 2016) traffic control (Adler et al., 2021; Hollander & Prashker, 2006; Staňková & Boudewijn, 2015; Bazzan & Klügl, 2014; Li, 2022), behavioral interactions (Elvik, 2014; Bazzan & Klügl, 2014; Li, 2022; Bensegghir et al., 2018; Zhang et al., 2010), routing operation (Sun et al., 2021) and transport safety (Meng et al., 2020) is summarized in Table 1. These classical game theoretic models possess several limitations, including rationality and selfishness (c.f. Section 2.3.2). Several behavioral economists and psychologists have criticized the basic behavioral assumptions of 'rationality and selfishness' for their inadequacy regarding human cognition and its limitations. Researchers have also questioned the rationality of transportation policies in the literature, even addressing the social influence others can exert on them. The issue of selfishness, however, is virtually untouched, as stated by Daniel McFadden, recognized as the "forefather" of travel behavior modeling: *"new results challenge the standard assumption of maximization of individualistic utility, indicating that social networks as information sources, reciprocity, and altruism enter human behavior and cannot be ignored"* (McFadden, 2013). Therefore, travelers are not necessarily selfish "Homo Economicus" species, but rather "Homo Sociologicus," a species characterized by cooperation and social interaction (Hirsch et al., 1987).

Recently, however, a new interest in executing these strategies has emerged due to advancements in evolutionary game theory (EGT). The research community has developed different evolutionary game models dealing with various features of the road transportation system. This article comprehensively reviews evolutionary game models advancing the road transportation network. As per the authors' knowledge, this is the first EGT-based review article covering various aspects of the road transportation network. The key contribution of this manuscript is as follows.

1. This manuscript presents a state-of-the-art review on EGT applied to the road transportation network.
2. The manuscript has divided the literature on EGT advancing the transportation network into four categories, i.e., choice-based analysis, traffic management, routing operation, and transport safety.
3. The choice-based analysis covers the detailed EGT-based advancements in route choice, mode choice, and lane selection. Traffic

management provides a detailed discussion on how EGT models have enhanced traffic control efficiency as well as the optimal congestion pricing mechanism. The routing operation includes the traffic/vehicle routing issues, route guidance, and driver's response. At the same time, transport safety applications include evolutionary game models dealing with optimal driving strategy, safety-aware speed recommendations, and collision/accident prediction.

4. Provides a comparative criticism of various evolutionary game models dealing with choice-based analysis, traffic management, routing operation, and transport safety.
5. Drawing upon the comparative evaluations, this work offers guidance for academic researchers, engineers, policymakers, and industry professionals by outlining potential research avenues within the field of road transportation networks.

The remainder of the manuscript is structured as follows; Section 2 presents an overview of game theoretic concepts and their classifications. Section 3 provides an in-depth review of evolutionary game models dealing with choice-based analysis, followed by traffic management modeling in Section 4. Section 5 summarizes the routing operation in the context of EGT. Transport safety is discussed in Section 6. Further, Section 7 provides the related challenges and future work, and finally, Section 8 concludes the work.

2. Game theory: concepts and classifications

Game theory (GT) is a branch of mathematics that studies competitive scenarios in which the actions of others substantially impact the outcomes of the participants. GT provides a framework for analyzing decision-making in practical situations, especially when dealing with intricate and interactive strategies, to facilitate optimal choices. The outcome of the game is influenced by various factors, including background information, environmental conditions, and interactions among individuals and groups. Strategic options can be chosen and implemented simultaneously, sequentially, once, or repeatedly, leading to corresponding outcomes or advantages (Abdoos, 2021).

2.1. GT concepts

The GT study focuses on how players utilize tactics to maximize their payoffs. A game solution gives alternate decision-making options and determines the game's conclusion. Games consist of several essential components, including players, actions, strategies, payoffs, and order. **Players** are individuals or groups who make choices to maximize their utility or objective function. Once the game rules are established, all participants must abide by them. Depending on the game type, players may have complete awareness, partial awareness, or no awareness of the actions taken by others. In many game systems, players have limited or no information about their counterparts (Piliz & Al-Fagih, 2019). In transportation networks, **actions** are the moves taken by players, such as drivers or authorities. **Strategies** are the players' action plans and rules. There are two types: pure strategy, where actions are chosen based on available information, and mixed strategy, where probabilities are assigned to each possible action (Camerer, 1991). **Payoff** represents the rewards obtained by participants based on their decisions. It is determined by the players' strategies or behavior. Utility functions are commonly used to express the payoffs associated with different strategy combinations. Players aim to maximize their utility by selecting suitable strategies (Lopez Mejia et al., 2022). **Order** in game theory refers to the sequence of strategic decisions made by players. The order of actions can lead to different outcomes, even with the same players and strategies. **Equilibrium** occurs when players employ their most effective strategies, resulting in a stable outcome. Nash Equilibrium (NE) is a solution concept in non-cooperative games where each player's strategy is the best response to others' strategies. In a NE, no player can improve their payoff by changing strategies unless others also change. A pure strategy

Table 1
Comparative study of the available game theoretic-based review articles dealing with the transportation network.

Transport objective		(Elvik, 2014)	(Zhu et al., 2022)	(Di & Liu, 2016)	(Ji & Levinson, 2020)	(Adler et al., 2021)	(Hollander & Prashker, 2006)	(Klein & Ben-Elia, 2016)	(Staňková & Boudewijn, 2015)	(Bazzan & Klügl, 2014)	(Li, 2022)	(Benseghir et al., 2018)	(Zhang et al., 2010)	(Sun et al., 2021)	(Meng et al., 2020)	This article
Travel behavior	Mode choice		x	x	x	X		x							x	X
	Route/Lane selection								x							X
Traffic control	Traffic signal	x														X
	Congestion pricing			x	x					X	X					X
Interaction	Signalized	x													x	X
	Non-signalized															X
Routing operation	Traveler-authorities						x			X		x				X
	Traffic routing															X
	Route guidance															X
	Drivers' response															X
Transport safety	Vehicular network												x			X
	Optimal														x	X
	Speed															X
	Driving strategy				x											X
Collision														x		X
No. of reviewed articles on EGT		0	0	0	5	2	0	3	0	0	1	0	6	0	1	235

NE is achieved when both players' best responses coincide. A mixed strategy NE involves a probability distribution over players' possible strategies (Pilz & Al-Fagih, 2019; Tushar et al., 2014).

2.2. GT classifications

GT encompasses classical and non-classical games. Classical games can be further classified as cooperative or non-cooperative, while non-classical games include matching and evolutionary games (Pilz & Al-Fagih, 2017). (GT) encompasses a wide range of games, including classical and non-classical games. Classical games can be further classified into two main categories: cooperative and non-cooperative games. In a cooperative game, players have the ability to establish enforceable agreements and form coalitions. This means that multiple players can join forces and work together towards a common objective. Within a coalition, participants can coordinate their strategies and negotiate how to distribute the overall payoff among themselves. The focus in cooperative games is on collaboration and collective decision-making.

On the other hand, non-cooperative games explore situations where players make decisions based on self-interest, typically without sharing information or forming binding agreements. Each player acts independently, striving to maximize their own individual outcome. In non-cooperative game theory, the emphasis lies in examining and modeling scenarios where each player's optimal choice is influenced by their beliefs about the actions and strategies of other players. Players are assumed to be rational decision-makers who seek to maximize their own gains. A key principle in non-cooperative game theory is that players should not make arbitrary assumptions about the actions of others. Instead, they should try to anticipate the moves of their opponents based on their understanding of the game's rules and the assumption of rationality. Rationality implies that players make logical and strategic choices, taking into account the available information and their objectives. This approach helps in predicting and analyzing the behavior of players in various strategic situations. (McCain, 2008). The non-cooperative GT is further categorized differently based on information, complete/incomplete information game, and perfect/imperfect information game. Based on Symmetry, non-cooperative GT can be classified as symmetric and asymmetric games. The games may be distinguished based on frequency as one-shot and repeated games. Based on forms, the games may be referred to as static (normal-form) and dynamic (extensive-form), and further, the non-cooperative GT can also be stochastic or deterministic (van Damme, 2014). The detailed illustration of the classifications in game theory can be found in Fig. 1 (Ahmad et al., 2023).

2.3. Evolutionary game theory

Evolutionary game theory (EGT) is a mathematical extension of game theory that incorporates frequency-dependent fitness and strategic aspects into evolutionary dynamics. EGT has gained increasing interest in various fields such as engineering, economics, sociology, anthropology, and social sciences in general, along with philosophy (Cressman & Apaloo, 2018).

2.3.1. Conceptual approach of egt

An evolutionary process consists of two main components: mutations, which introduce variation, and selections, which favor certain types over others. In evolutionary games, mutations play a key role according to evolutionary stability criteria, while choices are emphasized by replicator dynamics criteria (Adami et al., 2016).

The evolutionary stable strategy (ESS) is a strategy that can withstand evolutionary selection pressure (Brown & Wohletz, 1995). The criterion of evolutionary stability has notable points. Firstly, it does not question rationality. Secondly, it establishes a close connection between payoffs in the game and the stability of a population with a high-payoff incumbent strategy. These payoffs are directly related to biological "fitness," representing the expected number of surviving offspring. Thirdly, evolutionary stability focuses on a strategy's ability to withstand evolutionary pressure once it is reached, rather than how a population reaches that strategy. Thus, it does not add anything new to the concept of NE. Furthermore, evolutionary stability applies to human behavior in various situations, including economic interactions. (Cressman & Apaloo, 2018). Additionally, the criterion of evolutionary stability can be considered too strict. As a result, weaker criteria such as neutral stability (Smith, 1976) have been developed. In the case of an ESS, mutants are expected to earn less than the incumbents, ensuring robustness against equilibrium entrants. This condition ensures that no mutant can earn more than the incumbent strategy.

As defined by Maynard Smith (1974), an Evolutionary Stable Strategy (ESS) is considered dynamically stable if, when the values of the evolutionary parameters are perturbed, the force of natural selection will return to the ESS strategy, and the community strategy will deviate from the ESS strategy. However, in most classical works on evolutionary games, the dynamic stability of an ESS is not considered in relation to natural phenomena.

If a player's unilateral deviation from their equilibrium strategy results in a negative payoff, they are considered to be in a strict Nash Equilibrium (NE). While a strict NE captures one aspect of evolutionary stability, commonly referred to as a local optimum, it is widely recognized that a strict NE does not adequately capture the concept of

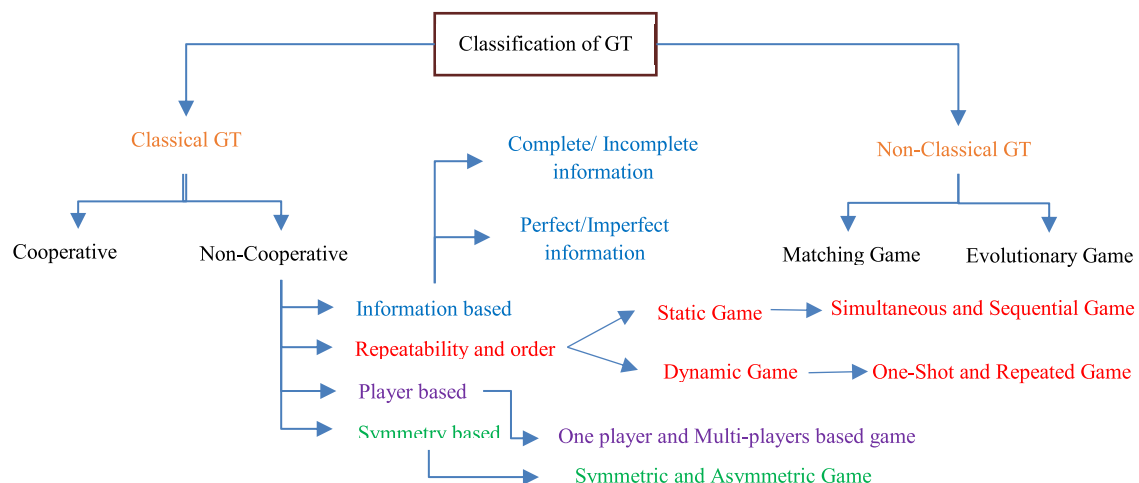


Fig. 1. A comprehensive GT classification (Ahmad et al., 2023; Pilz & Al-Fagih, 2019).

evolutionary stability. The Hawk-Dove game serves as an illustration of this phenomenon, where two players compete for a fixed resource value. Each individual adopts either the Hawk (k) strategy, acting aggressively until one is injured or the opponent submits, or the Dove (D) strategy, retreating as soon as the opponent becomes aggressive. In the context of the Hawk-Dove game, an ESS can be understood as a NE with an additional second-order stability criterion. This reinforces the concept of NE by acknowledging that while every game with a finite number of players and strategies may have at least one NE (assuming mixed strategies are allowed), not every game can be classified as having an ESS (Cressman & Apaloo, 2018).

2.3.2. EGT for transportation networks

Transportation systems can experience degradation due to unexpected events like adverse weather, accidents, natural disasters, and unpredictable human behavior. These events can have long-lasting effects, and the dynamic adjustments made by users impact the evolution of system performance and traffic flow patterns. To accurately measure the reliability of transportation networks, it is crucial to consider the dynamic interaction between user behavior and system performance. Classical GT has been widely used to address these issues (Alvarez et al., 2008), but achieving efficient performance remains a challenge. In this context, evolutionary game theory (EGT) has provided valuable insights into specific evolutionary questions and has gained popularity among social scientists as it addresses some limitations of classical GT. The following sections provide an overview of three key issues in this with classical GT.

2.3.2.1. Issues with equilibrium selection. A NE does not necessarily guarantee optimal rewards for each individual involved. This is exemplified in the prisoner's dilemma, where the only NE occurs when both agents defect (Arifovic, 2000).

However, there is a concern regarding the use of NE as a concept for determining game payoffs. When players are limited to using pure strategies, NE is not always guaranteed. While NE exists in non-cooperative games where players choose mixed strategies, its relevance to real-life situations has been questioned. In cases where rational agents are required to use only pure strategies (possibly due to high implementation costs associated with mixed tactics), it is important for game theorists to acknowledge that some games lack a feasible solution (Benseghir et al., 2018).

In certain games, multiple NE states can exist, posing a significant challenge when citing NE as a concept to determine optimality. It becomes difficult for a rational agent to choose among several possible equilibrium states. How can an agent determine which of the possible equilibria is the "best" to settle upon? Various attempts have been made to address this issue, each with some intuitive appeal. However, the concept of NE has undergone numerous refinements, indicating that even in games with multiple NE values, each equilibrium can be explained by a refinement already described in the literature. Consequently, the focus has shifted from selecting one of several NE solutions to choosing one of several refinements (Punniyamoorthy et al., 2021).

2.3.2.2. Issues with hyperrational agents. Classical GT relies on the assumption that players possess a high degree of rationality, which stems from utility theory's development—the theoretical foundation of GT. For instance, determining an individual's cardinal utility function typically assumes that the individual has well-defined preferences for various lottery outcomes based on their choices. This assumption requires each agent to have an infinite number of well-defined and consistent preferences due to the multitude of possible lottery outcomes (Weirich, 2018).

However, numerous experiments conducted in economics have demonstrated that these strong assumptions about rationality do not align with actual human behavior. Humans seldom exhibit the hyper-

rational behavior predicted by classical GT. Instances where individuals express preferences such as A preferred to B, B preferred to C, and C preferred to A are not uncommon, which contradicts the notion of consistent preferences. Moreover, experiments involving "beauty pageant" games have shown that many commonly held assumptions in games do not hold true. In contrast, evolutionary game theory (EGT) can explain specific behaviors observed in animals and insects without relying on strong assumptions of rationality, suggesting that rationality is not essential for GT analysis as previously believed. Consequently, EGT may provide better explanations and predictions of human choices by effectively accommodating weak assumptions of rationality. Notably, the framework introduced in Ganzfried and Sandholm (2010) offers a significant advantage in its ability to establish links between the micro-level behavior of individuals and the macro-level dynamics governing changes in the population (Bacharach, 1994).

2.3.2.3. Issues with dynamicity. Dynamic theories are undoubtedly more comprehensive, desirable, and acceptable. However, some studies suggest that attempting to construct dynamic theories without thoroughly understanding the static aspects would be futile. Evolution, regarded as a dynamic theory, explicitly models interactions between individuals in a population, as highlighted in the second approach to evolutionary game theory (EGT). In contrast, traditional game theories lack explicit treatment of the dynamics of rational deliberation. Thus, EGT fills a significant gap in classic game theory by explicitly examining the dynamics of rational decision-making (Sumaila & Apaloo, 2002).

While classical GT can try to capture some dynamics by modeling games in their extensive form rather than the normal form, it should be noted that games of reasonable complexity are often challenging to handle in this way. Additionally, traditional game theory presents a player's strategy as a predetermined specification of actions based on the information available within the game itself. This approach assumes hyperrational players and neglects the process of observing opponents' behavior, gaining insights through observation, and making adaptive moves based on learned information. Traditional game theory lacks the ability to model the dynamic element of gameplay. In contrast, EGT inherently incorporates dynamic factors, demonstrating its significant advantages in this regard (Sumaila & Apaloo, 2002; Gavidia-Calderon et al., 2020).

2.3.3. Applications of evolutionary game theory in sustainable transport network

The field of EGT has proven to be highly applicable across a range of disciplines, including economics, biology, politics, and computer science. Its versatility makes it particularly valuable for addressing complex problems. In the realm of transportation systems, EGT offers a comprehensive conceptual framework for modeling group interactions, making it well-suited to tackling transportation-related issues. EGT has been employed in numerous areas within the transportation domain, providing solutions to various challenges. These applications include traffic assignment, ridesharing matching, traffic signal control, route guidance, lane changing, congestion pricing, and more. By utilizing EGT, researchers and practitioners are able to analyze and optimize transportation systems in a way that considers the strategic behavior of multiple actors. One of the key strengths of EGT in transportation networks is its ability to adapt to changing factors over time. Transportation systems are dynamic and subject to evolving conditions, such as fluctuations in traffic demand, changes in infrastructure, and shifts in user preferences. EGT provides a framework to capture these dynamics and account for them in the analysis and design of transportation systems. For a comprehensive overview of how EGT is applied in the context of transportation networks, Fig. 2 serves as a valuable reference. It summarizes the various applications of EGT in this domain, providing insights into the diverse ways in which the theory is utilized to address transportation challenges.

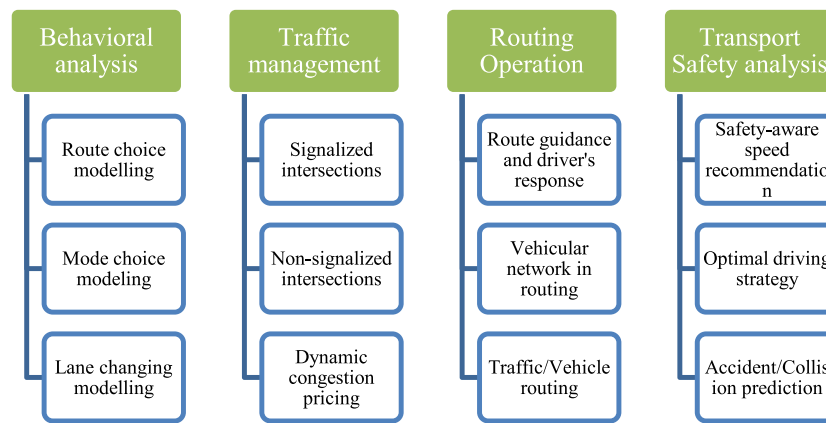


Fig. 2. Applications of EGT in the transportation network.

3. EGT and travel behavior analysis

The urgent need to transition towards a more sustainable transportation system stems from the far-reaching impact that travel choices have on various aspects of our lives and their potential consequences for the future. Governments and policymakers worldwide have recognized this pressing issue and placed it high on their agendas. Consequently, a significant amount of theoretical and empirical research has been dedicated to understanding the factors that shape travel choices and developing methods to influence these choices in order to achieve more sustainable mobility. Exploring the role of other individuals or agents in the decision-making process regarding travel choices has been a focal point across multiple disciplines, including sociology and economics (Bacharach, 1994). Researchers have sought to identify and analyze the complex interplay of socioeconomic, environmental, and cultural factors that significantly influence travelers' decision-making. These factors include travel costs, required travel time, associated satisfaction, dissatisfaction (e.g., waiting time), trip characteristics, and tour complexity. Among these factors, the value of travel time emerges as one of the most influential. The value of travel time pertains to the amount that passengers or drivers are willing to pay or can earn to avoid an increase in travel time when considering an average or typical mode of transportation (Madhuwanthi et al., 2016). This factor reflects the importance placed on minimizing travel time and indicates the trade-offs individuals are willing to make in terms of monetary or opportunity costs to reduce the time spent on their journeys.

As a result of recent research, it has become increasingly evident that the so-called "soft factors," such as perceptions and latent attitudes and social networks, play an important role in travel behavior (Huang et al., 2021). There is also literature on traffic behavior that considers the effect of other drivers, for example, the impact of a leader's speed or spacing on the behavior of a car driver (Huang et al., 2020). According to Dugundji and Walker (2005), the influence of the social environment on decision-making has been incorporated into empirical models in various ways, for example, via field variables. The systematic utility of a given decision-maker reference entity is determined by how many entities have made this choice or dummy variables are used to account for an individual's membership in a particular group. There are, however, some assumptions underlying these models. Thus, the GT setups can solve several choice-based transports-related problems. However, as explained in Section 2, the classical game theory has shown multiple limitations which EGT has overcome. This Section has covered an in-depth analysis of the EGT in travel choices such as route, mode, and lane selection.

3.1. Travel mode choice

One of the essential classical models of transport planning is the choice of mode of transportation. The primary reason is that public transit plays a crucial role in policy formulation. With the acceleration of industrialization throughout the world, growth rates are rising, incomes are increasing, and there is a high demand for mobility. It is known that as the city population increases, congestion and environmental problems result in traffic conditions that can cause delays and accidents that result in enormous economic loss every year due to increased traffic. Attracting passengers from private vehicles to mass transport methods appears to be a solution. However, it is not easy to achieve given that mass transport facilities provide them with a high level of comfort. Researchers have conducted studies to determine the relationship between mode choice and various factors influencing it to alleviate such deteriorating transportation conditions (Wu et al., 2015).

Typically, studies exploring the impact of situational factors on travel mode choice focus on modeling individual travel decisions. Conversely, the examination of psychological factors often involves analyzing aggregated travel mode choices, treating all circumstances as sources of error variations (Li et al., 2015). Therefore, to develop a realistic choice mode model, it is crucial to consider a range of influencing factors. Three distinct types of passengers can be identified based on their travel behavior: 1) individuals who consistently use public transportation irrespective of circumstances, 2) those who can adapt their travel mode based on specific situations, and 3) individuals who consistently rely on cars regardless of the circumstances (Zhang et al., 2017). By acknowledging and incorporating these different passenger types, a comprehensive and nuanced understanding of travel mode choices can be achieved.

3.1.1. Travel mode choice and game theory

When implementing traffic management policies, having an appropriate mode choice model is crucial. The Multinomial Logit model, based on the Logit concept developed by the researcher, is widely used for analyzing the mode choice process. However, the assumption of "independence of irrelevant alternatives" in multinomial logit models is not always rational and can be misleading. To address this limitation, nested logit models with a tree-branch structure have been introduced, offering some relief from the independence assumption. Nevertheless, these structures may not align with the principles of utility maximization or provide substantial improvements over the multinomial logit model (Li et al., 2020).

Alternative mode choice models, such as Probit (Havlíková & Zámečník, 2020) offer increased flexibility but are rarely utilized in mode choice modeling due to the additional constraints they impose on the covariance matrix. This results in challenging high-dimensional

integrals that need to be solved. On the other hand, mode choice models based on neural networks demonstrate higher accuracy as they can effectively capture complex relationships between the explanatory variables. However, their "black box" nature poses challenges in interpreting the results (Pineda-Jaramillo, 2019). Fuzzy logic-based mode choice models, which closely align with human linguistic expression, have also been explored. However, selecting the most appropriate membership function can be time-consuming and demanding (Ebrahimi & Bridgelall, 2021).

In the realm of statistical models used for travel mode choice, game theory (GT) algorithms have shown promising performance. GT algorithms do not make drastic assumptions about the data but instead learn to represent non-linear and complex relationships based on the available data. The researchers have conducted multiplayer-economic mode choice experiments to further investigate the application of GT algorithms in this context. (Chidambaram et al., 2014; Sunitiyoso et al., 2011; Ziegelmeyer et al., 2008; Innocenti et al., 2013; Rey et al., 2016). Overall, choosing an appropriate mode choice model is essential for effective traffic management policies. The Multinomial Logit model is commonly used, but alternative models such as nested logit, Probit, neural network-based, and fuzzy logic-based models offer different advantages and considerations. Additionally, game theory algorithms have demonstrated their potential in modeling travel mode choice, providing a valuable approach to capturing non-linear and complex relationships in the data.

However, these models could not evaluate the effect of measures other than economic ones and focused only on analyzing the impact of "pecuniary treatments" on private and public transit demand. As a result, the mode choice behavior is affected by some variables, including the type of information available regarding the trip and the individual's preferences. It must be adjusted repeatedly to become stable (Rey et al., 2016). In light of the above factor, in recent years, researchers have used the EGT to investigate research on how travel behavior evolves dynamically over time.

3.1.2. Mode choice from egt perspective: private and public travel mode

Travel behavior and the corresponding travel mode choice are often recurrent events rather than isolated incidents. In such cases, travelers aim to maximize their utility by continuously adapting their strategies (Calastri et al., 2019). While discrete choice models provide a framework for analyzing travel mode options, they primarily focus on static analyses and are unable to capture the dynamics of the decision-making process. To address this limitation, researchers have explored the integration of dynamic theory into the logit model, leading to the development of the concept of logit dynamics.

The logit dynamics concept offers an alternative approach that incorporates dynamic analysis into travel mode choice. Several studies have utilized logit dynamic differential equations to describe how travel modes evolve and are chosen over time (Qin et al., 2019; Rojas López & Wong, 2019). By applying the principles of logit dynamics and integrating them with concepts from EGT, researchers have developed models that depict the dynamic behavior of decision-makers in a game-theoretic framework.

This integration of logit dynamics and EGT allows for a more comprehensive understanding of the evolving nature of travel mode choice. It enables the exploration of how individuals adapt their decision-making strategies in response to changing circumstances, interactions with other decision-makers, and the overall dynamics of the transportation system. By considering the dynamic aspects of travel mode choice, researchers gain insights into the long-term patterns and dynamics of mode selection, providing valuable information for designing effective transportation policies and interventions.

3.1.2.1. Generic game model for private and public travel mode choice. EGT-based models offer valuable capabilities in evaluating the

effectiveness of policies through the analysis of payoff functions. These models allow for the inclusion of various variables, such as monetary considerations, time factors, and other relevant influences, within the payoff function. The choice of variables depends on the specific objectives of the model application. By employing payoff functions in EGT-based models, researchers can derive substantial value and insights (Qin et al., 2019; Rojas López & Wong, 2019).

The utilization of payoff functions in the analysis of EGT-based models enables a comprehensive assessment of policy outcomes. It allows decision-makers to quantitatively evaluate the impact of different policies or combinations of policies on desired objectives. The inclusion of monetary variables in the payoff function enables the assessment of economic benefits or costs associated with policy interventions. Time variables can provide insights into the efficiency and timeliness of policy implementation. Other influencing variables, specific to the context of the analysis, contribute to a more nuanced evaluation of policy effectiveness (Calastri et al., 2019).

Studies examining the application of EGT-based models have demonstrated the considerable value derived from the utilization of payoff functions. These functions enable researchers and policymakers to assess the potential benefits, trade-offs, and overall effectiveness of different policy scenarios. By quantifying the outcomes using appropriate variables and considering multiple dimensions of impact, such as monetary and time-related factors, EGT-based models enhance decision-making processes and provide a rigorous basis for policy evaluation and design (Hernández et al., 2018).

To provide a comprehensive explanation of the EGT-based model for mode choice, let's consider a road transportation network that encompasses both public and private travel modes, specifically buses and cars. In this bimodal road transportation system, the EGT-based model incorporates four key elements: group, pure strategy, mixed strategy, and payoff. The "group" refers to the total number of trips between origin-destination (OD) pairs, which generates the overall traffic demand denoted as D . The set of pure strategies, represented as $\{S_p = i, j\}$, includes the options available to travelers, distinguishing between the private travel mode (cars) denoted by i and the public travel mode (buses) denoted by j . In the EGT-based model, the mixed strategies are represented by the set of mode choice probabilities, denoted as $\{\rho_i, \rho_j\}$. These probabilities, ρ_i and ρ_j , reflect the likelihoods of selecting the private or public travel mode, respectively. The payoff function, associated with the negative of the travel cost, plays a vital role in the EGT-based model. It represents the evaluation of each strategy in terms of its utility or benefit. The goal is to minimize the travel cost, and the payoff function assists in determining the optimal mode choice. Fig. 3 provides an illustration of the mode choice probability process within the EGT

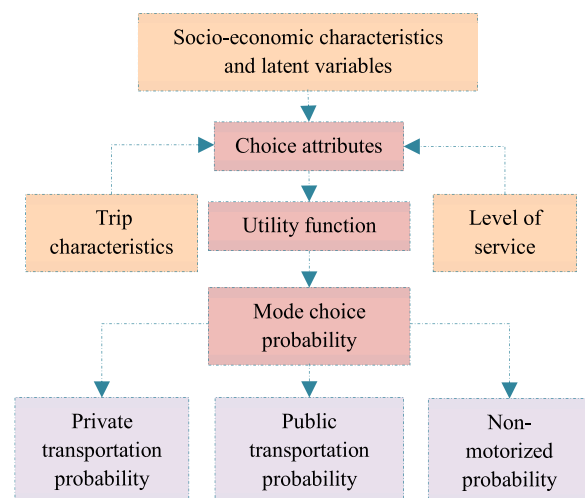


Fig. 3. Illustration of mode choice probability process under EGT model.

game modeling framework. It visually depicts how the probabilities of choosing different travel modes are determined based on the strategies and associated payoff values. By incorporating these elements, the EGT-based model for mode choice in a bimodal road transportation system offers a comprehensive framework for understanding and analyzing travelers' decisions. It enables the evaluation of mixed strategies, the calculation of mode choice probabilities, and the assessment of payoff values, all of which contribute to a better understanding of mode choice dynamics within the transportation network.

EGT considers that road travelers continuously change their choice of strategy. Due to this, it becomes essential to decide whether the dynamic process will eventually converge to the NE point. The dynamics of a group over time are generally described by a dynamic equation, as mentioned in Eq. (1). Where $\bar{p}$ denotes the mode choice probability derivative concerning time and $\mathcal{C}_{m,i}$ and $\mathcal{C}_{m,j}$ are the monetary cost of a car and bus.

$$\bar{p}_i = \mathcal{C}_{m,i}(\rho_i) \text{ and } \bar{p}_j = \mathcal{C}_{m,j}(\rho_j) \quad (1)$$

Suppose the traveler selects the private mode of transportation, denoted as option i. In this scenario, the probability of switching from the private mode to the public mode, represented as i to j, is referred to as the modification agreement and denoted as ρ_{ij} . Consequently, the dynamic equation can be expressed as Eq. (2).

$$\bar{p}_i = C_{m,i}(\rho_i) = \sum_{j \in S_p} \rho_{ji} \rho_{ji} - \rho_i \sum_{j \in S_p} \rho_{ij} \quad (2)$$

In reference to Eq. (2), the terms $\sum_{j \in S_p} \rho_{ji} \rho_{ji}$ and $\rho_i \sum_{j \in S_p} \rho_{ij}$ correspond to the proportion of participants adopting strategy i and the proportion of participants transitioning from strategy i to j, respectively. The specific form of ρ_{ij} determines the type of evolution dynamics observed. For instance, if the Replicator Dynamic (RD) model is employed to describe the dynamic choice process of travelers (Wu et al., 2020), the modifying agreement ρ_{ij} can be expressed as $\rho_j[E_j - E_i]_+$, and the RD model can be formulated as Eq. (3). Where $\bar{E} = \sum_{j \in S_p} \rho_j E_j$ or $\bar{E} = \sum_{i \in S_p} \rho_i E_i$, E_i, E_j denotes the payoff functions of mode i and mode j.

$$\bar{p}_i = \rho_i[E_i - \bar{E}] \quad \bar{p}_j = \rho_j[E_j - \bar{E}] \quad (3)$$

RD equation refers to the "rate of change" in mode choice probability is proportional to the "mode choice probability" and the "difference in an average and expected payoff."

3.1.2.2. Recent developments in private and public travel mode choice. Several research studies have employed the EGT model to investigate various aspects of travel behavior and the relationship between such behavior and influencing factors. One notable study by Levinson utilized an EGT-based model to explore the connection between travel costs and congestion. The findings of this study suggested a congestion policy that could potentially reduce expected travel costs (Levinson, 2005). Chen conducted a study focusing on adjusting mode choices in response to management policy objectives by manipulating equilibrium parameters. This approach aimed to enhance the effectiveness of managing urban transportation systems (Wu et al., 2020). The application of evolutionary games to model mode choice effacement strategies led to the conclusion that a game equilibrium corresponds to a stochastic user equilibrium (SUE) (Guan & Pu, 2010). This result was established through an examination conducted by Xiao and Wang, who developed an EGT model to explore how government regulations influence travel behavior. They emphasized the significant impact of subsidies for public transportation and the regulation of private vehicle use (X. & W., 2010). Similarly, another study investigated the influence of traffic enforcement on traffic violations using an evolutionary game model that encompassed both motor vehicles and non-motor vehicles (G. & D., 2013). The researchers in this study analyzed the impact of traffic

enforcement measures on promoting compliance with traffic regulations. Taking into account various travel behaviors, researchers in a different study examined the evolution of transportation from private to public modes of transportation (Sunitiyoso et al., 2011). This investigation considered multiple factors and dynamics involved in the transition. In order to measure the expected benefits resulting from different strategies, a dynamic game model of traffic flow was developed within the framework of EGT, incorporating prospects' values (Yang & Qian, 2014). It was revealed that players may not accurately estimate the utilities associated with different strategies. These studies demonstrate the diverse applications of EGT in the field of transportation research, shedding light on the intricate relationships between travel behavior, influencing factors, and management policies.

The EGT models discussed previously primarily rely on the Replicator Dynamics (RD) approach, where passengers adjust their travel patterns based on the system's average utility and their current traffic behavior. However, RD has faced criticism due to several limitations pointed out by researchers. Firstly, the assumption of RD being accurate in all scenarios does not always hold true. Furthermore, with a binary RD model, there is only one stable equilibrium point when certain constraints are met, while with a multiple RD model, there is no stable equilibrium point at all. Logit dynamics has been proposed as an alternative method for modeling the evolutionary game of mode selection in order to overcome the limitations associated with RD (Lin et al., 2017).

Logit Dynamic Evolutionary Games (LDEGs) utilize a probability measure to represent the population state, indicating how each individual within the population employs their strategies (Xue et al., 2018). This LDEG model combines the Logit model and Evolutionary Game Theory (EGT) by incorporating rational decision-making, where individuals adjust their strategies to maximize random utility based on their preferences and incomplete information (Chen et al., 2010). Within the framework of dynamic discrete choice, an agent's utility in LDEGs is influenced by various factors, including their previous and potential future choice utilities. Factors such as time and cost play a crucial role in shaping an agent's utility. By considering these elements, the LDEG model captures the dynamic nature of decision-making in a changing environment (Chen et al., 2010). Group decision-making is a crucial aspect of evolutionary games, and the LDEG model considers the impact of required travel time, total cost, and the decision-makers on individuals' utility functions. The LDEG model outperforms other dynamic-based discrete choice models by evaluating the proportion of groups in each option within a system and determining if it has reached a stable state, where the group proportions remain unchanged. Unlike dynamic discrete choice models, the LDEG model considers the interactions and influences of multiple groups, accounting for the dynamic relationships between them (Wu et al., 2015). Dynamic discrete choice models excel in predicting individual choices from a given set of options. However, they are limited in their ability to forecast the likelihood of decision-makers selecting specific options or the distribution of choices among different groups within the system. To address this, the Logit Dynamic Evolutionary Game (LDEG) model incorporates the concept of random utility maximization probabilities. In the LDEG model, it is assumed that traveler strategies undergo continuous adjustments during repeated games. This continuous adaptation process may lead to an equilibrium where there is no further change in the proportions of individuals using different modes of transportation. The LDEG model leverages the probabilities associated with random utility maximization to capture this dynamic adjustment process. One significant advantage of the LDEG model is its ability to capture the dynamic response of users to changes in travel costs throughout the day or even over the course of a week. It considers the dynamic nature of travel behavior as individuals react and adapt to fluctuations in travel costs. By accounting for these dynamics, the LDEG model provides insights into how users may shift their mode choices in response to changing circumstances and conditions (Chen et al., 2010).

In the DLEG model, travelers are believed to adapt their strategies for

selecting travel modes based on the principle of random utility maximization. Unlike the RD (Replicator Dynamics) model, the DLEG model takes into account individual preferences and incomplete information, making it a closer representation of actual decision-making behavior in mode selection. To determine the payoff function in this model, all commuters are assumed to be homogeneous, implying that they have similar characteristics and preferences in their mode choices. This assumption allows for the calculation of the payoff based on the utility derived from each selected mode. By incorporating individual preferences and the consideration of incomplete information, the DLEG model provides a more realistic approach to understanding how travelers make decisions when choosing a travel mode (Bai et al., 2019; Jing et al., 2020; Yang et al., 2020).

Researchers (Bai et al., 2019; Jing et al., 2020; Yang et al., 2020) have introduced LDEG models to analyze the mode choice between cars and railways during China's toll-free expressway program on holidays. This model enabled them to determine a globally stable equilibrium state, where the mode choices reached a state of balance. By incorporating the dynamics of individual decision-making and considering the impact of the toll-free expressway program, the LDEG model provided insights into the long-term mode preferences of travelers. The derived globally stable equilibrium state indicated a consistent distribution of mode choices between cars and railways, offering valuable information for policymakers and transportation planners in understanding travel behavior and making informed decisions related to the toll-free expressway program.

In Mou et al. (2021), the researchers introduced an innovative LDEG (Logit Dynamic Evolutionary Game) model that incorporated both empirical utility and random utility maximization principles of individuals. Their research focused on the introduction of a high-speed rail speed-up program that provided passengers with a choice of travel modes. The authors conducted a comparative analysis between logit dynamics and RD. They discovered that the RD model only had a single stable strategy when specific constraints were satisfied, whereas the logit dynamic model exhibited the only stable strategy during evolution. This comparison emphasized the advantages of using the logit dynamic approach in understanding the stability and evolution of strategies within the context of travel mode choices, particularly in relation to the high-speed rail speed-up program (Antonioni et al., 2015). LDEG, based on the random utility maximization principle, suggests that individual travelers adjust their behaviors while considering incomplete information and their preferences. However, Simon (Behavioral Model of Rational Choice, 2021) proposed that individuals tend to seek satisfactory solutions rather than globally optimal ones from the available options. As Krishnan (Lin et al., 2017) points out, an individual may change from being a "satisficer" to being a "utility maximizer" if one alternative appears more appealing than the other. A model for calculating an indifference threshold has been proposed by Krishnan, which represents the minimum perceptible difference between two alternatives. Several studies have explored the idea of random utility maximization and indifference thresholds (Lioukas, 1984; Cantillo et al., 2010; Wei et al., 2014; Wang et al., 2016). However, the dynamic evolution of travel behavior incorporating the concept of indifference thresholds has not been extensively examined. This suggests a need for further research to better understand the implications of integrating the indifference threshold concept into the dynamic evolution of travel behavior.

3.1.3. Mode choice from EGT perspective: private and car sharing

The use of private automobiles is one of the most popular modes of transportation worldwide. It is undeniable that private cars provide a wealth of comforts and facilities; however, they have also caused an increase in pollution, traffic congestion, and parking shortages, which have adversely affected the environment (Xu et al., 2020). The public transportation system is considered one of the world's most economic and environmentally friendly modes. However, because of its time and space constraints, it is often unable to provide flexibility and privacy to

its users. In the last few decades, there has been a push to provide new and innovative urban transport alternatives in order to reduce the number of private vehicles on the road and to promote multimodal mobility as a means to reduce vehicle ownership and private vehicle traffic. As a result, there will be a more diverse range of transportation options that are accessible, equitable, efficient, and safe. Carsharing has become widely accepted as a sustainable form of transportation, which has become a viable alternative to the use of private vehicles because it combines the characteristics of both private and collective transportation (Liang et al., 2021). Basically, carsharing works based on the idea that individuals gain the benefits of having a private car without having to worry about the costs and responsibilities of owning one (Amar & Basir, 2018).

In the current state of carsharing, there are four significant types of carsharing that are being considered, including (1) round trip carsharing, whereby users are required to return vehicles to the locations from which they were picked up; (2) one-way or point-to-point carsharing (launched in Austin, Texas in 2009) is a form of carsharing that allows users to pick up a vehicle from one location and leave it at another location near the destination, and if they require another car later on, they are able to pick it up at another location; (3) a peer-to-peer carsharing system (launched in Cambridge, Massachusetts in 2010) that enables individuals to share their privately owned cars with other members states that Fractional ownership is a model that allows individuals to co-lease or buys a car (launched in Stockholm in 2014) (Ferrero et al., 2018).

As indicated by earlier surveys (Julsrud & Farstad, 2020), the most important motivating factors for adopting the carsharing system are the convenient neighborhood location and reliable availability, as well as economic factors such as cost savings and the need for additional vehicles. These factors significantly influence the mindset of people whose income is middle to low, which constitutes most of the population in developing countries.

3.1.3.1. Generic egt modeling for car sharing. Evolutionary games consist of players, a strategy set, probabilities associated with each strategy, and payoff functions related to each player. The proportion of players with different strategies will remain stable in an evolutionary game once it reaches an evolutionarily stable state (Liu & Ma, 2008).

Several studies have been conducted on games played by one population (i.e., games where all travelers play the same role) and in which travelers use pure strategies. In most studies, it is assumed that all travelers are owners of private vehicles and are part of a transportation network with a total demand of N ($N = 1$). Car sharing and private cars are the two modes of transportation available to them. So, $\{S \in c, p\}$ is the pure strategy set for travelers, where p and c stands for a private and carsharing mode of travel. Travelers who choose car sharing have a probability of ρ , and travelers who choose private cars have a probability of $1 - \rho$. If travelers choose carsharing, their payoff function is F_c and if they choose private cars, it's F_p .

Generally, travelers' payoff functions correspond to their utility, which measures the time and costs they save as a result of their travels. There would be monetary and time costs associated with traveling. Carsharing travel time ($\tau_{t,c}$) is generally determined by the distance from your current location to the carsharing station (start point), the distance traveled in your car-sharing vehicle and the distance from the car-sharing station to the destination. Reservations of any returns of car-sharing vehicles may also be made online or by telephone. Similarly, if traveling by private automobile, the cost of travel ($\tau_{t,p}$) includes the time spent traveling from the current location to the parking lot (start point), the time spent traveling within the private vehicle, and the time spent traveling from the parking lot to the desired destination (endpoint). Car sharing companies charge a fee for their services, which is the total monetary cost associated with using their services. Usually, car-sharing charges are based on the number of kilometers driven and the number of

minutes used. In addition to fuel costs for travel, parking fees, insurance, cleaning and maintenance fees, and other similar costs, the total monetary travel costs associated with private cars include the cost of fuel. It is not usually considered in the total travel monetary cost to include the expense of the purchase and depreciation of a private car because these costs are usually treated as sunk costs (Cui & Levinson, 2021). In general, using carsharing can result in lower total monetary travel costs than using a private car for the same distance, i.e. ($m_c < m_p$).

Apart from the traditional cost and time commitments, there are also some benefits derived from a perception of these benefits based on personal beliefs and preferences, which are difficult to quantify directly. As an example, travelers might consider and convert the psychological benefits of safe, comfortable, reliable, and private travel into psychological rewards (Cui & Levinson, 2021). Additionally, symbolism plays a vital role in driving the choice between different modes of transport (Miller et al., 2016). Private car ownership is often regarded as a symbol of social status in modern society, and the psychological benefit of using a private car could also be considered a symbol of social status. The concept of car sharing contradicts the established norm of owning a car and the perception that owning a car should be a status symbol (Ferrero et al., 2018). As a result, in order for people to use carsharing to its full potential, they must abandon the symbols and meanings of personal ownership. We hypothesize that travelers' perceived benefit is driven by their driving experience, which determines all the aspects of travel, including safety, comfort, privacy, reliability, and status symbolism, that constitute the driving experience. Most car sharing organizations provide vehicles generally inferior to private cars in terms of reliability, privacy, safety, and comfort when considering their performance, configuration, and condition. Thus, this can be expressed as the driving experience benefit of car sharing (r_c) is lesser as compared to private cars (r_p).

It is also important to note that travelers have some externalities associated with their travels. Such as managing vehicle dispatch, maintenance, and after-sales services as a result of a growing number of carsharing users could result in a decline in the quality of service offered to carsharing users. Due to the limited capacity of car-sharing organizations to provide services, if more people use them, the number of services available to each traveler will decrease, resulting in a negative outcome for the individuals who use car-sharing. Suppose that α is the perceived benefit of the negative externality of carsharing that the travelers perceive. A higher proportion of the population choosing private cars as a mode of transportation would also negatively affect private car users. (Butler et al., 2020) report that these adverse effects include problems with parking, pollution, and health hazards. Let β be the amount of benefits the travelers perceive from the negative externality that is created when using private cars. Finally, the payoffs of the two strategies would be Eq. (4) and Eq. (5).

$$F_c = \alpha\rho + r_c - m_c - \tau_{t,c} \quad (4)$$

$$F_p = \beta(1 - \rho) + r_p - m_p - \tau_{t,p} \quad (5)$$

3.1.3.2. Recent developments in car sharing and public travel mode choice. Some studies have used psychological approaches to construct human behavioral models to determine the reasons people use car sharing. Various theories (such as the theory of planned behavior, the technology acceptance model, the unified theory of acceptance, and the use of technology) have been used to examine people's motives for using carsharing (Ferrero et al., 2018). The results of these studies provide insight into how to promote car-sharing use. However, these theories do not sufficiently explain why individuals switch from private cars to carshare vehicles. The transformation of travel modes is, in reality, a matter of choosing the suitable travel mode, which is generally guided by microeconomic theory's utility maximization framework (Santos, 2018). As a result of complex interactions among many factors, different modes of transportation may be used differently by individuals. Thus, in

order to promote carsharing use and develop a sustainable transportation system, it is critical to understand travelers' selection processes and mechanisms for using carsharing versus private cars.

Early studies evaluated and predicted people's preferences for and behavior regarding joining a carsharing organization using discrete choice models, such as logit and probit models (Hu & Creutzig, 2022; Tamboli et al., 2020; Webb, 2019). However, in light of the drawbacks of the classical methods, a combination of logit and probit models along with the EGT method is developed to overcome these problems. In EGT, the goal is to provide an explanation for equilibrium behavior based on dynamic variables. It is usually based on the approach that games with a finite number of strategies arise to evolutionary equilibria when dynamics become convergent to evolutionary equilibrium. In this approach, behavioral dynamics can be modeled using revision protocols to explain the decision-making processes of myopic individual agents (Licai et al., 2020).

An evolutionary game theory is based on two core concepts: ESS and evolutionary dynamics. In an evolutionary dynamic, agents can modify strategies based on revision protocols that describe how they adjust their initial choices of strategies as they play the recurrent game. Logit dynamics are the evolutionary dynamics in the evolutionary game model of logit dynamics. Evolutionarily stable strategies describe a population's equilibrium when changes are made during a given period in accordance with a revision protocol (Qiao & Wu, 2018).

Car sharing and private cars are both means of transportation that are similar and alternative to one another, as well as independent from one another. Private cars are safer, more comfortable, and eco-friendly than car sharing, which is more economical and environmentally friendly. Depending on the traveler's preferences, they could decide whether to travel with a car-sharing company or a private car. This suggests that travelers' behavior regarding the recurrent selection of the two modes in terms of the frequency with which they choose one mode over another is similar to the logit dynamic.

Considering the above discussion, some studies have combined logit dynamics with evolutionary game theory to develop logit dynamic evolutionary game models that can explain how decision makers behave in game systems in the context of dynamic evolution processes. Assuming that travelers continuously adapt their strategies during repeated games in order to maximize their utility, the logit dynamic evolutionary game is applied to the field of travel mode choice. Ultimately, this may result in a constant proportion of people using a given transportation mode so that the system can reach equilibrium. For the analysis of travelers' mode choice behavior between private cars and public transportation, (Zhang & Guan, 2019) developed an evolutionary game model based on the indifference threshold of the logit dynamic. (Zhang & Li, 2022) has integrated the EGT with logit dynamics aimed to construct a logit dynamic evolutionary game model to explain and analyze the dynamic processes by which travelers decide between owning a private car or taking part in a carsharing program. The psychological benefit of driving was also considered part of the travelers' payment functions in addition to the traditional time and monetary costs. Using mathematical analysis, the logit dynamic evolution model was found to have a unique global stable equilibrium point. The authors performed an experimental, numerical analysis on the evolutionary model of this study based on data derived from a simple interview survey conducted in Qingdao City to further explain the model's dynamic evolution.

The study by Bin Hariz et al. (2020) examined the effects of increased cost-sharing rates on users' costs in public transportation networks under the assumption that externalities are positive when the number of cost-sharing riders increases. A strong argument can be made that the experimental approach is an effective empirical tool for studying travel choice behavior, especially in scenarios that are difficult to control in practice, as well as for evaluating the impact of policy on such scenarios.

Further, another major concern of traffic engineers is the last mile issue, which concerns how to complete a long journey at its end. There

are various ways to choose including buses, taxis, and shared bikes, which all affect individuals' personal utility. (Li et al., 2019) started off by quantifying what it would cost users to use the service. Li et al. (2019) has developed an evolutionary game model based on the replication dynamic equation for selecting travel modes based on pedestrian perception and travel cost. A study of how the travel modes evolve in infinite populations was performed by calculating the equilibrium state of the system under different cost parameters. Furthermore, a complex network was used to evaluate the scene of finite populations. The results of these two studies demonstrate that there is a strong relationship between travel cost change and the proportion of the three modes of transportation once the system reaches a stable equilibrium state.

Although only a limited amount of literature exists on the dynamic processes that occur when travelers choose to use carsharing over private cars, these previous studies contain approaches and insights that can be used in future studies.

3.2. Lane changing modelling

The lane-changing (LC) behavior is more complex when compared with common driving behaviors such as changing the speed. Commonly, the LC behavior can be categorized as active and passive LC. The driver's action to achieve a better driving environment refers to active LC, while the passive LC refers to the need to change the lane, which generally occurs in the ramp area (Kim et al., 2021). The LC process includes three steps: the intention generation stage, the execution stage, and the state maintenance stage. In the first stage, considering the surrounding environment and habits, the driver generates the LC options to switch lanes and directions (left or right) and select the appropriate LC gap. The driver executes the actions on the steering wheel and accelerator to achieve the target lane smoothly during the execution process. During this stage, the main changes reflect the subject vehicle's driving state (the subject vehicle doesn't indicate any specific type, it can be any vehicle), like how fast it is going, how it accelerates, and how it changes latitudinally and longitudinally. In the third stage, the subject vehicle driver maintains stability in the target lane by adjusting the current speed and position of the vehicle through subtle operations. However, the surrounding vehicles also get impacted due to the actions taken by the subject vehicle (Jia et al., 2021).

Further, based on purpose, LC may be categorized as discretionary and mandatory LC models (i.e., DLCs and MLCs). In order to efficiently complete turns or avoid road blockages, vehicles must get into the target lane for MLC instead of improving driving conditions on the lane targeted for DLC. A relative trade-off in both MLC and DLC can be achieved by constructing a single utility function from both LC directions. Due to their diverse motives, DLC models have a more difficult time regulating traffic flow and reducing crash risks than other models. It is possible to achieve a more natural LC system using MLC and DLC in combination (Guo & Harmati, 2022). The research community has developed various models to solve challenges associated with LC; Fig. 4 shows a generic classification.

Lane changers are decision-makers, serving as the central unit of analysis within the LC decision and utility theory model. A problem with this approach is that no consideration is given to the reactions of passing vehicles. The lane changer and the driver in the adjacent lanes can be affected by one another during lane changes. Lane changers' behavior is partly influenced by the actions of the previous vehicle and the action of the vehicle moving in the adjacent lane. An LC maneuver affects the fellow's driving behavior and the leading vehicle's driver (R. Yao et al., 2021). Such algorithms can be created based on the analysis of interactions between players using GT or mathematical models.

3.3. Developments in game theory-based LC models

From various perspectives, researchers have used GT to study the problem of LC on highways. Known as a pioneer, Kita (Kita, 1999) was

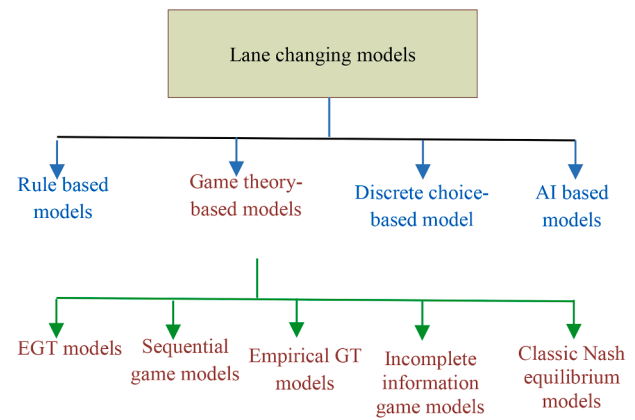


Fig. 4. Classification of lane-changing models.

the first to use a GT approach to analyze LC behaviors. As a result, the researchers' community investigated several factors, including traffic conflicts with drivers in interweaving areas (Arbis & Dixit, 2019), enhanced merging process (Liu et al., 2007), traffic safety (Guo & Harmati, 2022; A. Talebpour et al., 2015; Ali et al., 2019), and driving space requirements (Yu et al., 2018) in addition to comfort requirements (Wang et al., 2015). Additionally, new driving environments, such as the connected vehicle environment (A. Talebpour et al., 2015; Ali et al., 2019; Ali et al., 2021), as well as the automated driving environment (Na & Cole, 2019), are gradually being considered among LC players. As far as game models are concerned, researchers have applied pure strategy games (Kang & Rakha, 2017), repeated games (Kang & Rakha, 2020), mixed strategy games (Arbis & Dixit, 2019), Stackelberg games (Ji et al., 2021; Zhang, 2014), cooperative games (Li & Lin, 2021; Ladino & Wang, 2020; Zhao et al., 2021), EGT (Cortés-Berruero et al., 2016) and Harsanyi theory (A. Talebpour et al., 2015; Ali et al., 2019). Based on the various GT-based LC models developed in the last two decades, we have summarized their features in Table 3.

3.3.1. Generic lane-changing game model

To formulate a generic game model, two kinds of vehicles are considered; the first one is the subject vehicle (S_v) which intends to change the lane while the second one is the vehicle moving on the adjacent lane (A_v). The game between the driver of the subject ($S_{v,d}$) vehicle and adjacent vehicle driver ($A_{v,d}$) may be considered as two players game, such that players set P , i.e., $(P = \{S_{v,d}, A_{v,d}\})$. In the modeled game, the i^{th} player may choose an action a_i , $a_i \in A_i$ and $i \in \{S_{v,d}, A_{v,d}\}$, herein, the possible actions of $S_{v,d}$ might be "change lane or don't change lane," while $A_{v,d}$ may "give way or don't give way". Thus, the i^{th} player chooses a strategy s_i , $s_i \in S_i$ and $i \in \{S_{v,d}, A_{v,d}\}$, where various strategic combinations may be formed from the space of strategy profiles $S = X_{i \in \{S_{v,d}, A_{v,d}\}} S_i$. Consequently, the two players entail four strategy profiles, i.e., $S = \{(\text{Changing Lane, Give way}); (\text{Changing lane, don't give way}); (\text{don't change lane, give way}); (\text{Don't change lane, Don't give way})\}$. Suppose the payoff of the i^{th} is P_{off}^i , $i \in \{S_{v,d}, A_{v,d}\}$; specifically, the payoffs for $S_{v,d}$ and $A_{v,d}$ maybe denoted as $P_{off}^{S_{v,d}}$ and $P_{off}^{A_{v,d}}$, respectively. $A_{v,d}$ can also choose another lane, in addition to giving way or not giving way. But such an action would just be another game between competing vehicles. The strategic form game between $S_{v,d}$ and $A_{v,d}$ is tabulated in Table 4. In a strategy profile generated by pure strategies $P_{off,ij}^{S_{v,d}}$ and $P_{off,ij}^{A_{v,d}}$, $i, j \in \{1, 2\}$ are the individual payoffs belonging to $S_{v,d}$ and $A_{v,d}$. In the mixed-strategy game, with a certain probability the $S_{v,d}$ randomly chooses different strategies. Let's assume that the $S_{v,d}$ adopts a mixed strategy $(\sigma_{S_{v,d}} = (\rho, 1 - \rho))$, with a probability of ρ to change the lane and $(1 - \rho)$ for do not change the lane. Likewise, $A_{v,d}$ uses a mixed strategy $(\sigma_{A_{v,d}} = (\varphi, 1 - \varphi))$; that is, the $A_{v,d}$

Table 3

Comparative feature of EGT with different game models for LC.

Ref.	Models	Players	Features	Remark	Years
(Kita, 1999)	Empirical GT model	2	It modeled merging–giveaway conflicts in on-ramp merging sections by considering merging and the nearest through vehicles as a game. The model's results were corroborated with evidence from the field. It is a relatively simple model.	The model is reliable due to being calibrated from real scenarios	1999
(Liu et al., 2007)	Classic Nash equilibrium model	2	The model predicts how the driver will react during a complex merge maneuver. On-ramp merging vehicles join mainline traffic as soon as possible under safety constraints. In contrast, freeway on-coming vehicles aim to maintain their initial car-following state and minimize speed variations.	The model achieves the best personal choices with complete information	2007
(A. Talebpour et al., 2015)	Incomplete information game mode	2	The model has considered the flow of information under the characteristic of a connected environment. Transform the incomplete information game into an imperfect information game. More realistic than the gap-acceptance model.	The model separates lane changing into two types: mandatory and discretionary	2015
(Cortés-Berrueto et al., 2016)	EGT-based model	2	The impact of drivers' strategies on traffic flow is analyzed. Driver-driver interactions are incorporated, inheriting the two-lane traffic model's realistic dynamics. This game-theoretical formalisation facilitates LC behavior models and mobility performance monitoring.	The model promotes the cooperative rate in groups progressively	2016
(Yu et al., 2018)	Sequential game model	3	Through turn signals and lateral movements, the LC model mimics human behavior. Based on reactions, the model estimates the aggressiveness of surrounding vehicles/drivers. The developed model can extract the information and learn from interactions in real time.	The model supposes one reacts first and the other responds later	2018

Table 4Strategic and mixed strategic form game between subject ($S_{v,d}$) vehicle and adjacent vehicle driver ($A_{v,d}$).

	Strategic		Mix-strategic	
	$S_{v,d}$ Give way	Do not give way	$A_{v,d}$ Give way	Do not give way
Change lane	$P_{off,11}^{S_{v,d}}$ $P_{off,11}^{A_{v,d}}$	$P_{off,12}^{S_{v,d}}$ $P_{off,12}^{A_{v,d}}$	$\rho \cdot \varphi \cdot P_{off,11}^{S_{v,d}}, \rho \cdot \varphi \cdot P_{off,11}^{A_{v,d}}$	$\rho \cdot (1 - \varphi) \cdot P_{off,12}^{S_{v,d}}, \rho \cdot (1 - \varphi) \cdot P_{off,12}^{A_{v,d}}$
Do not change lane	$P_{off,21}^{S_{v,d}}$ $P_{off,21}^{A_{v,d}}$	$P_{off,22}^{S_{v,d}}$ $P_{off,22}^{A_{v,d}}$	$(1 - \rho) \cdot \varphi \cdot P_{off,21}^{S_{v,d}}, (1 - \rho) \cdot \varphi \cdot P_{off,21}^{A_{v,d}}$	$(1 - \rho) \cdot (1 - \varphi) \cdot P_{off,22}^{S_{v,d}}, (1 - \rho) \cdot (1 - \varphi) \cdot P_{off,22}^{A_{v,d}}$

gives way and doesn't give way to the $S_{v,d}$ with a probability of φ and $1 - \varphi$, respectively. In the mixed strategy game, the payoffs for $S_{v,d}$ and $A_{v,d}$ are tabulated in Table 4.

Payoff Formulations: As each driver ($S_{v,d}$ and $A_{v,d}$) changes lanes; they try to minimize changes in temporary driving status. At the same time, traffic participants typically make aggressive decisions in pursuit of travel efficiency as they move along. As a result, the safety of the drivers and the passengers may be at risk. Therefore, the payoff functions are determined mainly by considering time and safety. However, a more influencing factor may be incorporated dependence on the model formation need. Here, only time and safety are taken for readers' ease of understanding.

- (1) **Payoffs for Subject Vehicle:** During the lane change process, $S_{v,d}$ must consider the collision risk with the adjacent lane's vehicle. To avoid accidental collisions, both drivers must accelerate/decelerate accordingly. It begins to accelerate/decelerate when the $S_{v,d}$ starts to depart from his lane. Additionally, during the acceleration phase, $A_{v,d}$ may expect the subject vehicle to be the new leader and accelerate/decelerate appropriately. The risk of a collision can be avoided entirely if the $S_{v,d}$ does not change lanes, but this results in a delay. The payoff of the subject vehicle's decision to remain in the lane involves how much time each adjacent vehicle consumes to pass through the subject and how many passengers are in the vehicle. Table 5 shows the payoff for the $S_{v,d}$. In the table, A_{S_v} and A_{A_v} represents the acceleration for the subject vehicle and adjacent vehicle at the time of the LC process where the subject vehicle is considered as a new leader; S_{S_v} is the speed for the subject vehicle when it passes the adjacent

Table 5

Payoff for the subject and adjacent drivers.

	The payoff for the subject driver		The payoff for the adjacent driver	
	$S_{v,d}$ Give way	Do not give way	$A_{v,d}$ Give way	Do not give way
Change lane	$\delta_{11,0} + \delta_{11,1} \cdot A_{S_v}$	$\delta_{12,0} + \delta_{12,1} \cdot S_{S_v} + \delta_{12,2} \cdot A_{S_v}$	$\gamma_{11,0} + \gamma_{11,1} \cdot \tau$	$\gamma_{12,0} + \gamma_{12,1} \cdot A_{A_v} + \gamma_{12,2} \cdot S_{S_v}$
Do not change lane	$\delta_{21,0} + \delta_{21,1} \cdot \tau \cdot N_p$	$\delta_{22,0} + \delta_{22,1} \cdot \tau$	$\gamma_{21,0} + \gamma_{21,1} \cdot \tau$	$\gamma_{22,0} + \gamma_{22,1} \cdot A_{A_v} + \gamma_{22,2} \cdot S_{S_v}$

vehicle; τ is the consumed time for the adjacent vehicle when it passes the subject vehicle; N_p is the number of persons in the subject vehicle; and δ (a set of parameters need to be calibrated).

- (2) **Payoffs for Adjacent Vehicles:** Once the $A_{v,d}$ realizes that the subject vehicle is about to change its lane, two optional actions rest with $A_{v,d}$; give way or don't give way". If the $A_{v,d}$ gives way to the $S_{v,d}$, the $A_{v,d}$ will slow down the adjacent vehicle to avoid collisions with the subject vehicle. In such a case, the payoff is quantified by the time consumed during the lane change process. If the $A_{v,d}$ the doesn't give way to the subject vehicle, $A_{v,d}$ will speed up the adjacent vehicle. Considering safety as the utmost issue, the speed of the subject vehicle and the acceleration for the following adjacent vehicle must be taken into account.

The payoffs for $A_{v,d}$ are summarized in Table 5, where γ is the set of parameters that should be calibrated.

3.3.2. Reviews of existing EGT-based LC models

Relatively few studies have been carried out on the EGT-based LC models. The EGT applied to LC models provides a general explanation of the cooperative interactions between drivers from the perspective of an entire society. A two-lane road LC behavior rules are analyzed and formulated using evolutionary game theory (ZHOU et al., 2018). This paper simulates a vehicle road environment using the NetLogo simulation platform and provides acceleration/deceleration, speed keeping, and LC capabilities. Simulation is conducted to determine how traffic density, game rules, and driving behavior affect LC behaviors. Simulation results have shown that civilization directly impacts a vehicle's cooperative behavior and vehicle speed, traffic density, and traffic flow are directly related to cooperative driving behavior.

(Iwamura & Tanimoto, 2018) incorporated EGT into the cellular automation (CA) decision-making model, indicating that most of the applied mathematics based on the studied dilemma is traffic blockage. The authors in Iwamura and Tanimoto (2018) have estimated that the proportion of the group that utilizes a strategy will likely change over time as everyone strives to maximize their gains. Each driver applies maximization principles under high traffic density, resulting in a severe social dilemma. They developed a model that illustrates the dilemma effect. This model determines the appropriate gaps according to safety and incentive criteria, assuming that all vehicles will follow the rules derived from CA and replicate dynamically. Simulations show that it is sometimes impossible to achieve both social and Nash equilibrium simultaneously. Direct regulations must be established to minimize traffic density or prevent people from enjoying their maximum benefits by regulating traffic density. These measures encourage cooperative behavior. Similarly, the authors in Schmidt et al. (2021) have used a probabilistic CA model to update essential information. This assumed that every agent within a population could arrange itself to cooperate. During the experimental observation, the player who chooses the cooperative strategies pays the rewards other players receive. Moreover, the probabilities of collaboration between players are updated according to the five protocols proposed by Nowak (2006): Network Reciprocity, Direct or Indirect Reciprocity, and Group or Kin Selection. The simulation results prove that human cooperation behavior only occurs within a specific density range, so the traffic condition is the primary factor determining cooperative LC behavior. (Tanimoto, 2018) suggests that, along with using more than one reciprocity, two viscosity mechanisms can be combined, such as indirect and network reciprocity, to enhance cooperation over single reciprocity. However, the researchers concluded that integrated tools might negatively affect cooperative collaboration. Using simulators to develop cooperation strategies has been the subject of several recent studies (Zheng, 2014; Zimmermann et al., 2018). Further research should be conducted regarding the feasibility of weakening or eliminating dilemma effects.

The LC scenario involves two classes of driver-agents coexisting. The first maintains the lane assigned to them (cooperative strategy), while the second attempts to change lanes to progress (defective strategy). In phases of traffic flow resulting from bottlenecks caused by lane switching, (Nakata et al., 2010) identified several social-dilemma structures. (J. Tanimoto et al., 2014) examine another question: whether the formation of such structures can be achieved through regular lane changes without causing any explicit bottlenecks in the usual traffic flow. (J. Tanimoto et al., 2014) extended the work of (J. Tanimoto et al., 2014) for unusual traffic. As a result of the real-world traffic flow, which allows all drivers to switch lanes when necessary, it might be possible to classify drivers as well-behaved or ill-behaved. Cooperators are well-mannered drivers that switch lanes to overtake one another by moving into the fast lane for a brief time and then returning to the slow lane once they have overtaken one another. The word defector describes a dangerous driver who does not care whether the lane is fast or slow. Motivated by several ill-mannered drivers disregarding traffic rules pertaining to LC and maximum speed, (J. Tanimoto et al., 2014) extended the work of (J. Tanimoto et al., 2014) by establishing a model based on the revised S-NFS model coupled with EGT to investigate the possibility of social dilemmas arising from a frequent dangerous LC attitude during a typical traffic flow without explicit bottlenecks. (J. Tanimoto et al., 2014; J. Tanimoto et al., 2014) and (J. Tanimoto et al., 2014) found that for flow phases that are relatively high density, such as metastable phases and high-density phases, there is either a correlation between the structures and an n-person Prisoner's Dilemma (n-PD) game type or that these are similar to quasi-PD games. In such situations, heavy traffic jams result from ill-mannered drivers (drivers with defective strategies) who reduce the efficiency of social activities.

According to Tanimoto (2015), drivers' decisions regarding whether or not to change lanes in ordinal traffic flows evoke hidden social dilemma structures. A LC model was developed and applied with cyclic

boundary conditions to the EGT-integrated revised S-NFS cellular automaton model to confirm this. Interestingly, (J. Tanimoto et al., 2014) found that social dilemmas can only be observed in scenarios of medium vehicle density defined by the Prisoner's Dilemma game or variants. In these circumstances, data are scattered, corresponding to the metastable phase of the fundamental diagram. In this scenario, it would make sense since a driver is incentivized to change lanes when surrounded by other vehicles. In contrast, if all drivers adopt the same decision, the social efficiency of the system decreases tremendously, and there is a significant increase in traffic jams. Additionally, (Tanimoto, 2015) explored the relationship between dilemma strength and vehicle density.

(Iwamura & Tanimoto, 2018) discussed the interaction between social vehicles on the highway using evolutionary game theory and concluded that LC was associated with a social dilemma. Based on this fact, (Tanimoto et al., 2020) examined the behavior of social vehicles within various scenarios, including those in which vehicles are being driven by humans and those in which they are being driven autonomously. The mandatory lane changing process for bus exiting, compared to the lane changing procedure for social vehicles, has some distinct characteristics, such as higher traffic delays and discomfort caused by excessive acceleration. Taking into account the unique attributes of mandatory lane changes for bus exits is further addressed in R. Yao et al. (2021) to improve the accuracy of the models. Various scenarios were analyzed in R. Yao et al. (2021) for mandatory lane changes for bus exits. It demonstrated how buses and social vehicles interact according to evolutionary game theory. Observation and experience are constantly exchanged by bus and social vehicle populations for learning and optimizing decisions. Consequently, stable-state analysis and numerical experiments are used to investigate populations' evolutionary mechanisms of strategies. Further, (Chen et al., 2016) proposed a lane change formation simulation model with cellular automata to simulate uncontrolled mid-block crosswalks where pedestrians and vehicles interact. It is proposed to consider the three-second rule and modify the pedestrian rule set to resolve the deadlock among mixed flows by taking into account the right-moving preference of pedestrians. Real data collected at Jian She First Road in Wuhan, China, was used to calibrate and validate the developed model. This study demonstrates that the proposed model can simulate vehicle-pedestrian interactions and the formation of lane changes.

Using the revised S-NFS model, (Tanimoto & An, 2019) has developed an integrated EGT Cellular Automata traffic model that incorporates the density of traffic in the next 50 m for each vehicle and a decision-making process to determine whether a lane change should be attempted to minimize meaningless lane changes. The system is designed to provide information about traffic density within 100 m of the focal vehicle as part of a protocol for improving traffic efficiency on-premises. Based on systematic simulations, the presented lane change protocol significantly improves traffic flux compared to the conventional incentive and safe criteria-based lane change rules. As a result of (Tanimoto & An, 2019), such protocols to limit meaningless lane changes can be technical easily integrated into automated cruising systems that support both IoT and ITS in the future. In conventional traffic flows, social dilemmas are inevitable and can be relaxed with such protocols.

Motivated by Tanimoto and An (2019), the authors of (Sueyoshi et al., 2022) have proposed a new model of EGT integrated cellular automata based on the revised S-NFS model, where the agents have their maximum velocity distributed according to their level of mobility, similar to a real traffic flow field consisting of compact vehicles, trucks, and buses. (Sueyoshi et al., 2022) computes flow characteristics based on systematic simulations of multi-agent flows in the default system (maximum velocity is constant), mixed systems (maximum velocity is distributed), and correlated-mixed systems. Defectors tend to use higher maximal velocity strategies, whereas cooperators tend to use lower maximal velocity strategies. In addition to discussing the underlying game class underpinning each traffic flow system, (Sueyoshi et al., 2022)

quantitatively examined the social efficiency deficit, which measures how expansive a dilemma may be for each flow system discussed. As (Sueyoshi et al., 2022) explicitly discussed, evolutionary game theory (e.g., (Guo et al., 2019; Hassan et al., 2014; Zheng, 2014; Tanimoto, 2015; Tanimoto et al., 2020; R. Yao et al., 2021; Tanimoto & An, 2019) as recent studies) is a pretty powerful tool for analyzing traffic flow from social dilemma point of view.

4. Traffic congestion management

Congestion is a term used to describe a situation of overcrowding, overburdening, or overfilling that creates obstacles or hindrances. In the context of transportation, congestion refers to the condition where numerous vehicles are present, moving slowly and in an irregular manner (Pi et al., 2021). Within the transportation system, there are two primary types of traffic congestion: recurrent and non-recurrent. Non-recurrent congestion occurs sporadically due to disruptions in the normal flow of traffic, which can be caused by various factors such as accidents, natural disasters, work zones, and road construction sites (Cantillo et al., 2010). On the other hand, recurrent congestion refers to a scenario in which traffic congestion happens repeatedly or in a cyclical manner. Recurrent congestion is influenced by factors such as peak travel hours, complex network structures and plans, and frequent on-off ramping (Villalobos et al., 2008; Bui et al., 2017). Understanding the distinction between recurrent and non-recurrent congestion is essential for effectively managing and mitigating traffic congestion issues in transportation systems. By identifying the causes and patterns of congestion, transportation planners and authorities can implement strategies to alleviate congestion, improve traffic flow, and enhance overall system efficiency.

The following factors demonstrate that traffic congestion adversely affects the community's economy, environmental quality, and quality of life. (a) A rise in transportation costs, specifically those related to road and freight tasks, has a negative impact on national productivity and competitiveness. The Global Traffic Scorecard (2019) reveals that, on average, drivers in the United States lose 99 h a year due to traffic, representing \$1377 per driver. (b) A consequence of the increase in idle time for vehicles is an increase in CO₂ emissions. (c) Public dissatisfaction with the inadequate management of traffic on the highways as travel times to destinations increase (Morrow, 2008). Congestion impacts not only drivers but also individuals using public transportation, particularly those from lower socioeconomic backgrounds, who experience slower travel speeds and higher fares as a result of congestion. Therefore, alleviating traffic congestion is crucial for enhancing quality of life by minimizing travel delays, reducing transportation expenses, and enhancing overall network efficiency (Villalobos et al., 2008; Bui et al., 2017).

Congestion relief reduces traffic congestion by diverting traffic to less congested parts of the highways. In relieving traffic congestion, various factors such as traffic flow, density, and speed must be considered. The number of vehicles on the road at any given time is traffic flow, while average speed is considered at the place of speed. Traffic density is defined as the number of vehicles per unit of average speed. Many methods have been employed to resolve traffic congestion issues. In addition to optimal traffic signal control (TSC), turning restriction, traffic routing, and congestion pricing, there are other strategies. This section will discuss the most recent developments in those areas. Because this review focuses on approaches that integrate EGT approaches in traffic congestion relief, it encompasses the methods mentioned above.

4.1. Traffic signal control

Traffic jams in urban areas are becoming a growing problem that requires innovative control strategies, such as traffic signal coordination. The most common control system used in such strategies is a

central traffic-responsive controller, and as more traffic elements are involved, such as traffic lights, detectors, specialized hardware, etc., the more difficult it becomes to implement and maintain. Interoperability between these elements is another issue, particularly when obtained from different manufacturers (Gartner et al., 1980).

Based on their control strategies, TSC methods can be classified into three kinds: fixed-time, adaptive, and actuated. This classification was made primarily due to the types of data and algorithms used in optimizing traffic signal plans (Diakaki et al., 2002).

Fixed time TSC strategies are generally suitable for traffic signals that are relatively stable and consistent in traffic flow. This strategy uses offline optimization algorithms for TSC based on previously observed traffic data and ends up with a predetermined cycle length and split setting plan. By minimizing the average delay, (Thunig et al., 2019) developed the initial model for fixed-time traffic control strategies. Due to the highly dynamic nature of urban traffic systems, minor disruptions, such as traffic collisions, construction works, etc., can significantly change traffic volumes and adversely affect a predetermined traffic signal program.

As time progressed, visual systems became increasingly popular. In actuated control strategies, sensors are also utilized, and the controller determines the cycle length based on historical information. The adaptive strategy refers to modified forms of actuated control, which predict traffic conditions in real-time based on the data that is available at the moment (Hunt et al., 1981). Alternatively, an adaptive signal possesses the capability of adapting to traffic changes without the limitations associated with actuated signals. A key objective of adaptive control strategies is to optimize the traffic signal control plan in accordance with the current traffic conditions in each phase of the signal control process. Due to the fact that traffic signal plans must be regulated instantly, these strategies are not able to carry out as much detailed analysis as fixed-time strategies (Hassan et al., 2014). The actuated control method is responsive to traffic. Still, it is not designed to deal with fluctuating traffic levels, rendering it less than ideal, particularly in highly saturated traffic conditions. TSC requires a high degree of adaptability in order to adjust to different traffic conditions due to the highly dynamic nature of traffic phenomena. The TSC research problems can therefore be solved using a variety of methods. A wide range of methods is available, including (i) traffic theory-based methods, (ii) simulation-based methods, and (iii) data-driven methods (Sermpis et al., 2010).

4.1.1. Classical algorithms and software for traffic signal control

In order to optimize traffic signal controllers offline, several well-established algorithms have been developed, including TRANSYT (Meggio et al., 1979), which generates fixed-time operation plans that are optimized based on optimal coordination and timing criteria. The main disadvantage of TRANSYT is that it relies on historical data to compute plans for a static situation. In contrast to TRANSYT, the Split Cycle Offset Optimization Technique (Hunt et al., 1981) uses data from traffic-responsive buried detectors (called SCOOT). In addition to SCOOT, Sydney Coordinated Adaptive Traffic System (SCATS) also uses real-time data in order to improve traffic flow in Sydney. One of the main differences between SCOOT and SCATS is that one is a hierarchical system, while the other is not (Lowrie, 1982). The OPAC (Gartner, 1983), Prodyn (Henry et al., 1983), and UTOPIA (Di Taranto, 1989) are similar frameworks, but all of them are decentralized adaptive programs. These formulas do not consider explicit splits, cycles, or offsets.

In Prodyn, for example, a decision is made every five seconds regarding phase changes. Each intersection is typically set up to simulate every possible scenario based on what is detected in the areas next to it. From one intersection to the next, this information propagates with decreasing weight from one intersection to the next. There is the potential for both the complexity of the computation and the communication system to increase the implementation costs. A traffic-responsive urban traffic control system (Diakaki et al., 2002) was developed with the intention of managing networks on a large scale. The coordination

strategy involves making adjustments to the split, cycle, or offset (either all or some of them) as well as prioritizing the use of public transportation. Results of a simulation of morning peaks are presented in two different scenarios (small network, real-world) in two different studies. Comparing the performance of the experiment to a situation where no time synchronization occurred, the result was positive. Alternatively, the computation is centralized, and conflicts are resolved either by an expert in traffic management or through predetermined priority rules, in contrast to a multiagent system.

In view of these issues, it is evident that simple offline or even online optimization of traffic synchronizations in arterials is insufficient to deal with changes in traffic patterns. It is due to the fact that traffic is a dynamic process, and as a result, it is almost impossible to determine the optimal signal plan beforehand. A more flexible and robust approach is required in light of the increasing traffic volume.

Microeconomics is typically based on decentralized supply and demand and no centralized authority. A utility theory is used in this modeling to represent each market participant's preferences and perform operations regarding the utility of all choices. A decision theory is also utilized to deal with uncertainty and perform operations regarding each preference's utility. In contrast, a game theory is employed to deal with conflicting interests between market participants. The idea of game theory can be considered an extension of the concept of decision theory in the context of multiple decision-makers in a system.

In Gartner et al. (1980), Gartner et al. proposed that road users and traffic managers might have different objectives, each of which tries to maximize. To analyze this idea, (Fisk, 1984) compared it with the concept of games. To solve congestion at intersections, Alvarez et al. (Alvarez et al., 2007) proposed an approach based on noncooperative games. Cheng et al. (Cheng et al., 2006) developed a parallel coordinated approach to signaling, which he named CoSIGN, which stands for coordinated signals. The Nash equilibria for the game were computed using an advanced iterative process called sampled fictitious play. Each decision maker was considered a player who controlled a period of time for a signal during the game. The algorithm was tested on 75 signalized intersections in Troy, MI, demonstrating its robust scalability to realistic networks.

Similarly, some studies based on classical game theory have been conducted; however, they are subject to rationality and other multiple limitations, well explained in Section (2.2.2). Regarding human beings, it is less evident whether the same theory can be applied to players without intellectual abilities. Thus, EGT comes into the picture.

4.1.2. EGT perspective of traffic control at intersections

There is an essential distinction between problems associated with road-oriented traffic control and those related to vehicle-oriented traffic control. Road-oriented traffic control systems are based on road-based measures that manage traffic using external road-based interventions at fixed locations. These include signalized traffic systems incorporating traffic signals, message signs, etc. In vehicle-oriented traffic control, traffic is controlled through internal signals generated by the vehicle, such as advanced driver assistance systems. It also deals with non-signalized traffic systems (Eom & Kim, 2020). The current section reviews and discusses traffic control strategies based on the signalized and non-signalized taxonomy that primarily uses EGT approaches. Fig. 5 and 6 represent the four phases and players in the intersection game.

4.1.2.1. EGT-based traffic control at signalized intersections. Applying EGT to TSC systems is a relatively straightforward process for several reasons. First, the TSC system is a complicated nonlinear stochastic system. The optimum setting of traffic signals is, therefore, difficult to determine. The EGT allows for the possibility of obtaining optimal or suboptimal solutions. The second reason is that, in most traffic signal optimization methods, mathematical models used to estimate traffic flow are precise and accurate. The modeling of dynamic traffic,

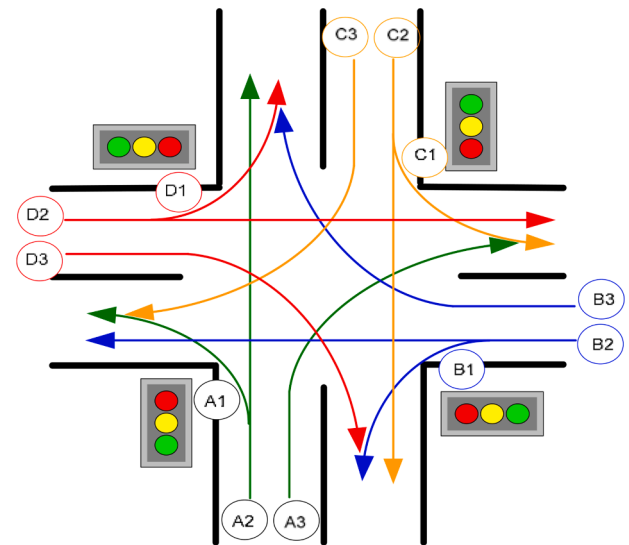


Fig. 5. A generic four-phase of traffic signal (Al-Khateeb & Johari, 2008).

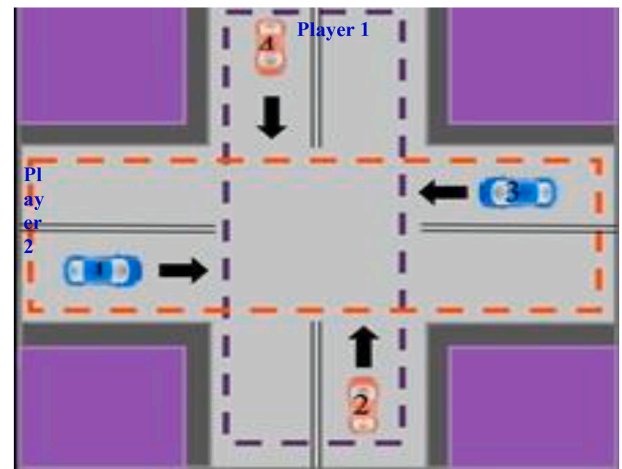


Fig. 6. Game Illustration at the signalized intersection (Elhenawy et al., 2015).

however, is very challenging. Lastly, EGT can be viewed as a broader study of mechanisms that facilitate or enable learning behaviors under complex, uncertain, and changing conditions (Cressman & Apaloo, 2018). Thus, EGT is a suitable approach for studying traffic systems with dynamic characteristics. Researchers have conducted a few works on the applications of EGT in the field of TSC.

A study by Taale et al. (Taale et al., 1998) compares evolutionary algorithms (an evolution strategy) with a typical Dutch traffic light controller (the RWS C-controller) to develop a traffic light controller for a single simulated intersection. For both systems, the results were comparable. However, due to the complexity of the dynamics of such network traffic nodes, they could not test their system on multiple coupled intersections. Such networks' learning or control mechanisms could result in additional exciting behaviors. Using EGT's idea and selection mechanism, (Li & Yu, 2012) examines a new signal light control mechanism for the complex and random traffic flow at single intersections, which aims to minimize the mean traffic flow delay. In order to evaluate the proposed method, the traffic simulation software VISSIM is used to evaluate its performance; following that, the effect of the proposed methodology on control is compared to the fixed-time control method and genetic algorithm-based signal control. Simulation results demonstrate that the proposed approach reduces traffic flow mean delays and improves signal control effectiveness. Using the concept of EGT,

the authors in Xia (2018) have proposed a model that utilizes chicken game decision-making for timing traffic signals at two adjacent intersections. Intersections were considered game players, while queue lengths were considered payoff functions. Based on the chicken game, Nash equilibrium was obtained for mixed strategies. The queue length of each phase can be significantly reduced compared to other non-coordinated methods like fixed timing control. Comparing the values of vehicle travel times and topping numbers at each arterial section, the simulation results showed that the game-based mechanism was more efficient. The proposed method allows multiple intersections to be scheduled simultaneously with distributed control. According to De Oliveira et al. (2005), the authors have developed a method of solving the traffic light coordination problem using EGTs that incorporates both autonomous and implicit coordination and a classical centralized model. The optimal Asynchronous Partial Overlay (OptAPO) algorithm for cooperative mediation is used to formulate the problem as an online optimization problem. This paper achieves two objectives: it tests the OptAPO algorithm in dynamic environments, which was not explicitly designed for dynamic environments, and it proposes a new method for synchronizing traffic lights without requiring assistance from traffic experts.

Autonomous coordination depends very much on the learning process and abilities. An agent-based system is one of the most commonly used models for studying the interaction behavior between vehicle to vehicle and vehicle to infrastructures, i.e., multiple individuals. As a consequence of their direct impact on one another, the interaction process between different agents is more complex. Agents' interests are often at odds with each other regarding improving their interests. As a general framework, EGT provides an account of how individual fitness is related to a population's proportion of strategies (Wu et al., 2021; Bazzan, 1996; Bazzan, 1997). By summarizing the above, EGT assumes that agents' objectives must be compatible with appropriate evolutionary dynamics rather than a matter of logic. The behavior of self-interested individuals cannot be predicted by evolutionary models. In complex environments where agents interact with each other, it describes how they can make decisions. The agents must be able to adapt to the non-stationarity of their environments in these complex environments.

Fig. 7 illustrates the game learning model for multiagent systems. The entire game process begins with initializing relevant parameters for each game agent in the group, followed by the commencement of a game between the agents in accordance with specific matching rules. As a result of observing its income, the income of the others, its own choice,

and the choice of the others, the agent learns and finds a solution according to a certain learning algorithm. In the final stage, the opponent is chosen again according to certain guidelines, and the process is repeated until a final evolutionary equilibrium is reached.

In light of the above discussion, (Wu et al., 2021) proposes a mathematical model for studying game-based (EGT) learning technologies in multiagent systems. In this model, agents are described as learning from the competition and adjusting their behavior independently based on their own interests through local interaction. The control of the overall performance of a multiagent system is achieved through the choice of the appropriate game type, the design of the appropriate income calculation method, and the modification of the evolution rules. Evolution does not require the specification of how a particular individual behaves; therefore, a dynamic behavior for each individual is not necessary. The overall goal can only be achieved through the adaptive evolution of the population. It is a controlled, intelligent, and self-adaptive adjustment process. A novel EGT-based analysis has been presented in In Bazzan (1996) in order to coordinate individually motivated agents working on two intersecting roads, aiming for synchronization of signals. In a long-term interaction, agents coordinate toward a specific strategy to allow traffic to flow more smoothly on the road by selecting signal plans. A particular advantage of this approach is its ability to deal with interactions with fully motivated individuals. Since agents only need to be aware of their own rewards, not those of their adversaries, coordination can be modeled with minimal or no communication requirements. Another advantage of this approach is that complex n-player games can also be modeled as a series of two-player games to accommodate their complexity.

Further, (Bazzan, 1997) has proposed EGT integrated with Multi-Agent Systems to show the optimal coordination of signalized TSC. Since traffic signal control involves frequent changes in traffic conditions, the proposed approach proved efficient since a coordinated system based on an explicit negotiation framework would require a great deal of communication that is not feasible in real life. A further advantage of this approach is that it is flexible enough to allow agents to achieve an equilibrium without being stranded in it even if nature alters traffic patterns in the meantime. Their adaptability allows them to adjust to the changes in traffic conditions and reach the best coordination state. As an extension of the approach, it would be interesting to examine how neighbors communicate their intentions, especially when an agent receives two communications with contradictory demands. Based on the reinforcements received, agents must determine which neighbor has proved trustworthy, which one has made a greater contribution, and so forth. In Bazzan (2005), Bazzan developed a distributed approach to coordinate TSC agents using EGT. In a two-person game, the agents competed against each other in an attempt to reach Nash equilibrium according to evolutionary stable strategy. This approach did not require agents to know the strategies of their opponents, thus resulting in a low communication burden, improving the reliability of the method. Various scenarios in the experiment were observed that showed different performances of the decentralized coordination process (e.g. if there are nearly equal amounts of vehicles flowing in opposing directions, one direction has a significantly higher flow, etc.). It is therefore the manager's responsibility to facilitate once he has recognized the scenario and has taken the appropriate action.

(Bazzan, 1997; Bazzan, 2005) possess three advantages: no central agent is required to decide the direction of coordination; agents can dynamically design traffic light coordination subgroups depending on current needs and traffic volume in a given direction; thus, agents do not have to communicate explicitly to determine where to coordinate traffic lights. The system designer, however, must explicitly formalize payoff matrices (or at least the utility and preference levels of the agents) to make these figures available. The approach can therefore be time-consuming in situations where there are a number of options for coordination (for example, all four directions: south, north, east, and west) and/or complex traffic networks (for example, several transversal

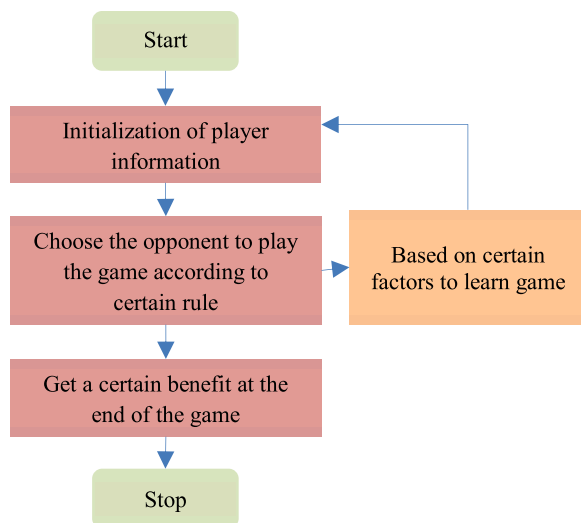


Fig. 7. Game learning model for the multiagent system to achieve evolutionary equilibrium.

and parallel streets need to be considered in addition to the main arterial).

In Anantsuksomsri and Tontisirin (2016), traffic congestion is analyzed from a micro behavioral perspective using an EGT-based theoretical framework. To analyze the behavior of road users in a setting of roads in the downtown Bangkok business district, the author extends the conceptual framework of EPD. This analysis indicates that the actions taken by agents during travel are influenced by the costs associated with travel delays, traffic accidents, and the fines levied for traffic violations. According to this paper, motorists intentionally violate traffic regulations - a form of negligence - because they believe they will be able to save travel time by doing so. However, this belief can result in even more congestion if enough motorists share it. In comparison with scenarios in which more motorists behave responsibly, the average traffic speed is much slower when all motorists act irresponsibly - or always defect. Therefore, congestion is a result of strategic time-saving measures. Based on the study's results, it can be concluded that losing context preservation could cause the extinction of cooperative behavior, which counters the finding of the earlier studies. Thus, this theoretical framework offers a perspective for analyzing collective strategic decisions regarding using a common resource from the perspective of EGT.

The TSC is the subject of multiple studies. It has, however, rarely been investigated what happens when a driver and a traffic light use evolutionary mechanisms simultaneously, particularly in the absence of a central control system. In some instances, if co-evolution is carried out in a decentralized manner, then chaotic or auto-organized situations may occur as a result of co-evolution. The reason for this is that, although ITS has reached a high level of technical sophistication, it is unclear how drivers will react to these systems. Accordingly, (Van Zuylen & Taale, 2004) has constructed a bi-level three-player game with the functionality to control traffic lights and assign routes, which simply means that both traffic light control and route selection must be considered when playing the game. Since all players are assumed to have complete information, all players are expected to be aware of each other's movements. It is important to note that even though the paper reports intriguing conclusions, for example, the advantage of cooperation among the players, these conclusions are probably only applicable to that simple scenario. It isn't easy to justify the assumption that drivers always take their shortest routes in a real-world application. Thus, in Bazzan et al. (2007), the authors discuss how to route choice decisions (by drivers) and traffic light control measures (by traffic signals) are influenced by co-evolving decision-making. Especially in large-scale situations (e.g., hundreds of drivers), the results show that all actors co-evolve to improve travel time and occupancy. Results were compared to those obtained in scenarios where traffic lights or drivers evolve and those obtained with a centralized optimization method in various scenarios. There are two major directions in which this work can be extended. The first approach is to integrate the supply and demand simulation tools separately developed by the authors, which are more user-friendly and provide the means of simulating more realistic scenarios as travel becomes more complex. A second extension is introducing heuristics in reinforcement learning algorithms to improve the performance of MAS algorithms. Since reinforcement learning is a difficult problem in non-stationary environments, especially with multiple agents involved, this is not a trivial extension.

Numerous studies conducted in the literature analyze the two parties involved, i.e., the driver and the traffic signal operator, from a static or dynamic perspective. A few studies have investigated the interaction behavior of traffic operators and vehicle drivers when pedestrians are present, as they cross signals at various intersections, which may be considered a multi-party multi-group interaction behavior. The mix of traffic increasingly impacts urban traffic operations flowing through the cities. In order to detect the mechanism, one should examine the properties of pedestrians, their psychology, and the characteristics of their behavior.

To study the evolution law of multi-group behaviors in a pedestrian

crossing, (Gou et al., 2016) has constructed a bounded rationality-based EGT model to analyze "Chinese style road crossing" of pedestrians under the influence of motorized vehicles and traffic operators. The stability evolution strategy of adventure and normal pedestrians was analyzed based on Lyapunov differential equations. In the analysis of a stable evolution strategy, it has been revealed that adventure pedestrians can choose either to use the rules according to their personal preferences or to violate them. A normal pedestrian can evolve into a pedestrian who obeys the rules or even evolves into a pedestrian who mimics the behavior of a pedestrian who breaks the rules in an attempt to stop behavior that is breaking the rules. In addition to favorable factors, several limitation factors affect adventure pedestrians. Results indicated that adventure pedestrians were subject to path dependence, but this was dependent on the profits of normal pedestrians. In Zhang et al. (2021), a multi-group behavior evolution-based EGT game model is developed for pedestrian crossing behavior. The concerned process was triggered by the perception of risk and the behavior choices made by multiple groups. It discusses evolution trends, evolutionary stability strategies, and factors that affect multi-group behavior's evolutionary path. According to this study, vehicle drivers, pedestrians, and traffic system operators managed their evolutionary strategy equilibrium not only in accordance with their own earning potential but also in accordance with that of their neighboring groups. A revenue factor and a limiting factor affect the behavior of the traffic participants, and the evolution process conducted under the influence of these two factors. Traffic managers, vehicle drivers, and pedestrians were influenced by the behavioral evolution strategies developed by each group, as well as those of the other two groups. As a further measure of understanding cyclists' traffic violations and enforcement strategies, (Tang et al., 2020) proposes an EGT analysis method. A fixed penalty strategy, as well as a dynamic penalty strategy, was employed to analyze evolutionary equilibrium stability. In simulation-based numerical experiments, it has been demonstrated that; (i) the proposed method is capable of identifying the interactions between enforcement strategies and traffic violations; (ii) when traffic violations are perceived to have fewer benefits than the penalties associated with them, as well as enforcement costs being lower than enforcement benefits, the probability of cyclists committing traffic violations can be reduced by a fixed penalty strategy as well as a dynamic penalty strategy, provided that the dynamic penalty strategy provides law enforcement with a stable enforcement outcome, compared to a fixed penalty strategy. An enforcement strategy's effectiveness in reducing cyclists' traffic violations is influenced by a number of factors, such as how much penalties are imposed, how effective the enforcement is, how much it costs, and attitudinal factors (perceived relative benefits, public image costs, and cyclist attitudes toward risk).

There has been a significant improvement in the understanding of traffic intersections due to the above-discussed research, which has resulted in improved traffic control at signalized intersections. Non-signalized intersections behave very differently than signalized intersections since they lack such signal control mechanisms.

4.1.2.2. EGT-based traffic control at non-signalized intersections. It has been observed that traffic flows at non-signalized intersections play a significant role in determining the capacity of road traffic, particularly in urban areas. Secondly, non-signalized intersections are generally more common than traffic light intersections (Bouderba & Moussa, 2019), which leads to an increase in car accidents. Unsignalized intersection accidents typically result from a combination of factors, ranging from human factors to mechanical failures of vehicles to road conditions. Managing road users' conflicts is challenging without consideration of the human factor, especially when interacting drivers are intoxicated, fatigued, inexperienced, or distracted (Zhao & Jia, 2015). Understanding traffic dynamics in an organized environment is essential to improving traffic efficiency at an unsignalized intersection. Identifying the optimal strategies to solve the crossing conflict problem at such

an intersection is also necessary. A game theoretical framework can be used to explain this situation. Fig. 8 shows a generic game formation at a non-signalized intersection. A black car stands for a cooperator and a pink car for a defector in an intersection cell marked by an X.

Various games have been used to study non-signalized interactions, such as the evolutionary prisoner's dilemma or the Snowdrift game (also called the chicken game or hawk-dove game). Players can choose between cooperating or defecting in these two game settings. Using a predefined cooperation rate (a proportion of drivers who cooperate among themselves), Zhang analyzed traffic flow at a non-signalized intersection (Zhang et al., 2012). According to the study, traffic density and cooperation rate significantly impact traffic flow and energy consumption. According to J. Tanimoto et al. (2014), prisoner dilemma games occur in meta-stable, high-density traffic flow phases with relatively high densities.

A fixed proportion of cooperators and defectors is used in the abovementioned studies. Although real-life scenarios vary quite a bit, in the majority of cases, the driver learns from their experience and adjusts their strategy over time. Therefore, it cannot be fixed how much cooperation there is between drivers, which makes it necessary to discuss the temporal learning mechanisms, and how the drivers make decisions in response to the learning they have received (Zhao & Jia, 2015). It is commonly referred to as the strategy update rule (SUR) in many studies of EGT. Several SURs have been applied in EGT to optimize the TCS in recent years, including best-response, imitation, tit-for-tat, and reinforcement learning. An overview of SURs can be found in Qi et al. (Qi et al., 2015). A tit-for-tat SUR strategy involves cooperating with the opponent in the first round but following the opponent's lead in subsequent rounds. Taking the best possible strategy as a response to the current distribution of strategies of their competitors is considered to be the best response strategy. Unlike reinforcement learning, in which agents tend to repeat strategies that have shown satisfactory rewards in the past, imitation is based on the assumption that humans act similarly to how most people would react (Zhao & Jia, 2015). Different models have been developed based on these assumptions of learning the traffic mechanisms. Using computer simulations and experiments, Qi et al. (Qi et al., 2015) investigated the snowdrift game at non-signalized intersections. Through reinforcement learning, Zhao et al. (Zhao & Jia, 2015) investigated the impact of an uncontrolled intersection on traffic flow. In Payssan et al. (Paissan & Abramson, 2013), a grid network traffic game with unsignalized intersections has been developed that incorporates an imitation dynamic. Depending on how each driver perceives the strategy of the other drivers at the crossroad, the probability of each driver changing their strategy is derived.

To study the crossing conflicts at unsignalized intersections, (Bouderba & Moussa, 2019) employs a cellular automata approach along

with evolutionary game dilemmas. The dynamics of vehicles at the intersection are modeled using a set of different rules in which drivers employ different behavioral strategies. Numerous countries with poor traffic education have experienced this situation, from which imitation behavior, "do as the others do," usually develops. Hence, (Bouderba & Moussa, 2019) observed that, in a context such as this, the introduction of a group of well-educated drivers who adhere to the law, and an array of uneducated drivers who do not, results in an intersection game similar to a prisoner's dilemma, whose outcomes affect the efficiency and collective performance of the system. A study in Paissan and Abramson (2013) indicates that imitating drivers could change their strategies depending on the frequency with which they encountered other drivers who displayed the same strategy. In Bouderba and Moussa (2019), however, imitation is determined by the occurrence of an accident at the intersection; therefore, aggressive drivers will pay increased attention to crossing the intersection at subsequent times if an accident occurs (which occurs near the intersection). There may be a change in strategy on the part of these drivers so that they can become cooperative drivers. The cautious driver may imitate the aggressive driver, even if there is no accident, to cross the intersection more aggressively in the future. Thus, cooperating drivers may transform into defecting drivers through a change in strategy. The simulation results demonstrate that the game's outcome is independent of how many injectors are injected or the configuration of the vehicles on the road at the beginning.

(Yao et al., 2018) studies traffic flow at non-signalized intersections on a grid road. Iterative snowdrift games are played between the drivers in the network. An analysis of the traffic flow characteristics and the evolution of the drivers' behavior is conducted using a cellular automata model. To update the rules, the drivers employ the best-response strategy. The study reveals three significant findings. First, simulations show that cooperation rates fall with an increase in cost to benefit ratio, and this drop can be deduced analytically as a function of the cost-to-benefit ratio. Secondly, the authors observe that higher cooperation rates are associated with higher average speeds, lower densities, and more significant flows. The evidence suggests that defectors impede traffic flow at non-signalized intersections. Lastly, when the density is low, the drivers have less opportunity to update their strategies, so there is more randomness in the system.

In Chen et al. (2016), an EGT and cumulative prospect theory approach is used to model pedestrian-vehicle interactions at unsignalized midblock intersections. Under bounded rationality and risk, EGT models pedestrian behavioral differences in a group setting. However, evolutionary game frameworks are often incorporated into micro-simulation platforms that include numerous calibration parameters. As pedestrian and vehicle interaction models become more sophisticated, the evolutionary game framework becomes increasingly challenging to generalize.

4.2. Congestion pricing

The concept of congestion pricing has gained increasing scholarly and practical attention as an economic instrument for transportation demand management. Since the initial exploration of road pricing, theoretical and applied literature on congestion pricing has grown rapidly. The detail of road price is classified in Fig. 9 (AbuLibdeh, 2017).

Moreover, when considering optimal approaches, the congestion pricing issue can be categorized according to four criteria: (1) the utilization of first-best or second-best pricing strategies, (2) the application of static or dynamic traffic assignment techniques, (3) the consideration of homogeneous or heterogeneous user groups, and (4) the incorporation of deterministic or stochastic parameters. Congestion pricing models predominantly rely on the principles of static traffic assignment theory, which provides a foundation for analyzing and optimizing traffic flows under different pricing schemes (Tikoudis et al., 2015; Han et al., 2021; Han et al., 2017).

It should be noted that static models do not account for the time-

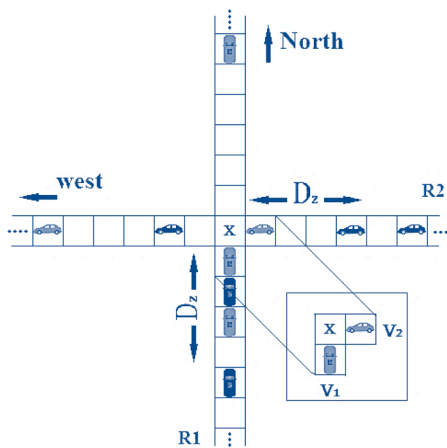


Fig. 8. Illustration of a game at a non-signalized intersection (Bouderba & Moussa, 2019).

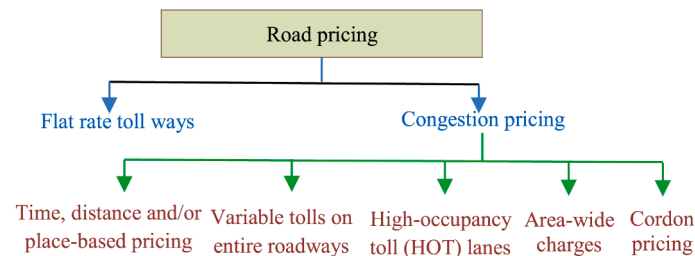


Fig. 9. Road pricing classifications.

dependent nature of traffic flows and travel behavior but only for the data flow in a non-variational manner. Moreover, there is an argument that classical congestion pricing models do not adequately account for the effect of tolls on future traffic congestion (Zheng et al., 2012). Given the inherent uncertainties, variability, and time-dependent nature of transportation systems, it is more logical and advantageous to employ dynamic congestion pricing (DCP) models. These models take into account the dynamic nature of traffic networks and are more effective in addressing the congestion pricing problem. Within the realm of DCP, two classifications exist based on the timescale distribution: within-day and day-to-day DCP models. Within-day DCP models focus on link flow patterns that vary throughout the day, with tolls adjusted accordingly. These models utilize the framework of dynamic user equilibrium (DUE) and system optima to achieve optimal spatial and temporal utilization of the transportation network. By considering the dynamic nature of traffic patterns and toll adjustments, within-day DCP models offer a more comprehensive and efficient approach to managing congestion (Saharan et al., 2020).

The dynamic transportation network considers the impact of departure times on congestion tolls, recognizing that congestion levels vary throughout the day. This concept acknowledges that the timing of trips can influence the effectiveness of congestion pricing measures (Richard et al., 1991). However, the application of this approach is often limited to simpler network structures and does not adequately address computational challenges. To address these limitations, researchers have developed more comprehensive models for complex transportation networks. In one study (Carey & Srinivasan, 1993), approximate analytical expressions are proposed for congestion tolls based on the Kuhn-Tucker optimality condition. This approach aims to capture the realistic characteristics of network congestion and tolling mechanisms. However, a key assumption in this model is that the demand is known in advance, and the optimal toll is determined based on these deterministic values. This assumption may not accurately represent real-world scenarios where demand fluctuates and is subject to uncertainty.

In practice, using deterministic and inelastic demand values can lead to suboptimal system performance. If the actual demand significantly differs from the assumed values, the congestion pricing scheme may not achieve the desired outcomes. Moreover, obtaining precise travel predictions in real-life situations is challenging due to various factors such as specification errors and limited availability of data. (Lu & Tettamanti, 2021). These uncertainties further highlight the importance of robust and flexible congestion pricing models. To illustrate the economic rationale behind transitioning from an equilibrium without tolls to a congestion pricing equilibrium, Fig. 10 provides a visual representation. This figure demonstrates the potential benefits of implementing congestion pricing measures, showing how the equilibrium conditions and societal welfare can be improved through appropriate tolling strategies.

Moreover, in a study conducted by researchers (Do Chung et al., 2012), a robust optimization approach was utilized to address the challenges of congestion pricing. One significant challenge in implementing robust congestion pricing is the uncertainty associated with

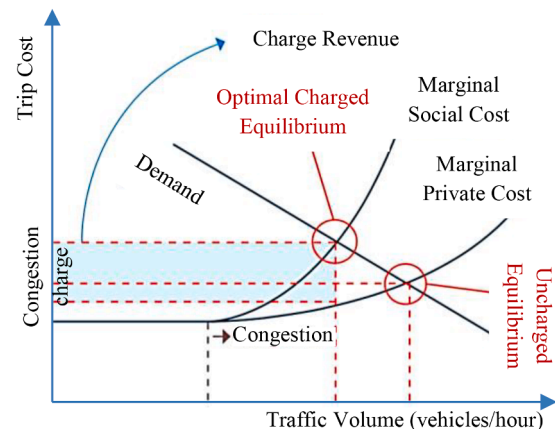


Fig. 10. Economic rationale for moving from untolled equilibrium to congestion pricing equilibrium.

user demand, which is not accounted for in traditional user equilibrium conditions. To overcome this limitation, a bi-level formulation was employed as a mathematical framework. This formulation incorporates equilibrium constraints and expresses the user equilibrium condition as a variational inequality, effectively capturing the range of possible demand scenarios. By considering a family of constraints, this approach accommodates the uncertainty in user demand and allows for more robust and reliable congestion pricing strategies.

These previous studies have overlooked the strategic behavior of travelers, particularly in situations where the outcome is influenced by the choices of multiple individuals. To address this, game theory has been widely employed as an analytical tool to capture the strategic interactions between players aiming to maximize their own utility in a given scenario. Various theoretical models, including the Prisoner's Dilemma, Social planner game, Cournot game, and Evolutionary game, have contributed to a deeper understanding of this process (Pilz & Al-Fagih, 2017). It is important to note that achieving a pure NE for traffic congestion pricing is generally not feasible. However, according to Ohazulike et al. (2012), under specific conditions, a pure NE can exist. This mechanism allows for the determination of the optimal toll through cooperative games among different actors. Game theory also offers insights into the bi-level nature of dynamic toll pricing problems, where drivers and traffic authorities are recognized as acting players. At the lower level, drivers select their routes to minimize travel costs without considering the impact on other network users. Meanwhile, at the upper level, road managers may introduce congestion pricing to minimize the total system cost, aiming for a User Equilibrium pattern towards a System Optimum (Shirazi et al., 2017). Nevertheless, classical game theory has inherent limitations, including assumptions of rationality (as discussed in Section 2.3). Therefore, the research community has shown a growing interest in employing EGT for dynamic congestion pricing, which provides a more nuanced approach to understanding and addressing these complexities.

4.2.1. Review of existing literature on EGT-based congestion pricing

Based on the EGT, (Sandholm, 1997) analyzes potential games with continuous player sets, including a general network congestion model as a special case. The authors concisely describe both Nash equilibrium sets, and equilibria sets resilient to small perturbations in aggregate behavior. The evolution of equilibria is well justified by a strong evolutionary argument. It was shown that evolution always increases aggregate efficiency in situations in which stable equilibrium is socially efficient. The results lead to constructing an evolutionary model that yields socially optimal behavior under parameterized congestion tolls. In conclusion, the authors demonstrate how continuous-player potential games can be characterized by demonstrating that their generalization would satisfy the anonymity condition as a generalization of finite-player potential games.

(Sandholm, 2001) has studied a class of games characterized by externality symmetry conditions, which occur when there are continuous player sets of players. Examples of these games include random matching games, congestion games, and games with expected payoffs. Under the broad class of evolutionary dynamics, the author has introduced a simple description of locally stable equilibria and shows that from all possible initial conditions, behavior tends to converge to the Nash equilibrium. Finally, the authors prove that the games studied here are limited cases of convergent sequences involving multi-players with a finite number of participants.

(Sandholm, 2002) has developed an EGT mechanism for implementing congestion pricing, showing that when users follow a myopic adjustment process (identifying the most effective paths and integrating real-time information), it eventually consolidates to steady-state behavior. As described in Sandholm (2002), price schemes can solve implementation problems in deterministic models with continuous populations and negative externalities. An efficient behavior can be rendered globally stable through deterministic adjustment processes. In order to prove these results, a concave relationship between the total payoff from externalities and the aggregate behavior of the population must exist. As a result of this condition, the price scheme generates global equilibrium conditions that are stable, efficient, and unique. The concavity condition, however, is specific to models of negative externalities, and when it fails, multiple equilibrium conditions are typically generated by price schemes. The authors in Sandholm (2007) have introduced a Pigouvian pricing method for congestion pricing and stochastic evolutionary implementation that take into account the limitations of (Sandholm, 2002). The proposed model considers either positive or negative externalities, or a combination of both, depending on the case. Therefore, the authors do not use concavity conditions but specify behavior stochastically. By choosing one of the many equilibria generated by the price scheme, one can recursively select one among the multiple alternative equilibria, enabling us to render the optimal behavior a long-term prediction. However, the proof of (Pilz & Al-Fagih, 2019) relies on a special property of the logit choice rule: when noise is low, the stationary distribution generated by this rule places a large proportion of mass on the maximal potential. It is essential to know how this specification influences the results. In addition, (Sandholm, 2007) is subject to another criticism regarding the length of time required for the predictions underlying it to become relevant. The expected time before departure from a stochastically unstable equilibrium can grow exponentially with the size of the population in evolutionary models. This may lead to doubt regarding predictions of play based on stochastic stability when a large population is present. However, when agents live geographically separated from one another and tend to interact with one another, this waiting time critique has little force. Once a stochastically stable strategy gains a foothold in one area, contagion dynamics may cause it to spread rapidly throughout the population. Consequently, Blume's selection theorem is still valid when applied to local interaction models (Kollman, 2018). The fact that long waiting times are far less of an issue in local interaction models gives us hope that the pricing mechanisms reviewed here may be relevant to local interaction models.

A key question for future research is how evolutionary implementation results will be achieved in such settings.

Drawing upon the examination of traffic congestion as a crucial factor, researchers in Wu and Li (2006) proposed a perspective that treats travel as a form of gaming process influenced by other activities. Employing the principles of evolutionary dynamics game theory, they conducted analyses considering both non-elastic and elastic demand conditions. The findings indicate that, with an acceptable dynamic framework for the congestion game and the introduction of a reasonable variable price scheme based on evolutionary dynamics, the game can converge to a unique Nash equilibrium. This approach allows for the attainment of an efficient state and optimization of society's driving choices. The variable price scheme employed embodies the quality of traffic congestion pricing, distinguishing it from price schemes based solely on marginal cost considerations.

A multi-agent integrated EGT system based on the day-to-day dynamics of traffic flow under tradable bottleneck permits (Wada et al., 2008) is proposed that minimizes the total transportation cost through decentralized agent behavior. To alleviate traffic congestion, a road manager imposes congestion charges on all users of the city's street network in accordance with Sandholm's congestion pricing model (Sandholm, 2002). According to this approach, the congestion tolls are determined solely based on the temporal flow of traffic, which is an evolutionary approach to implementing the model.

Users' information acquisition rates are taken into account in the models mentioned above of dynamic congestion pricing, which rely on hard assumptions. In most dynamic models, particularly those based on optimal control theory, an adaptive process is identified, modeled, and analyzed in relation to a learning mechanism without making any assumptions regarding the behavior of the users.

Considering the above limitations, the authors of (Dimitriou & Stathopoulos, 2008) propose an EGT framework for assessing congestion pricing strategies based on marginal travel costs. By taking into account the behavioral characteristics of multiple user classes, the framework is capable of identifying the dynamic impact of congestion pricing on the system's performance. This framework builds on tactics for variable road congestion pricing based on EGT and stochastic optimal control. Using a hypothetical network as a basis for analyzing the framework's fundamental characteristics enables the analysis of its essential characteristics. Moreover, the model is applied to a realistic network representing a part of the Central Business Area in Athens, Greece, which provides valuable insights into its features.

(Dimitriou & Tsekeris, 2009) presents a dynamic congestion pricing model based on EGT for urban roads. In this model, users' stochasticity, reliability, and heterogeneity of route selection are taken into account based on the value of travel time (VoTT) and the capability of obtaining real-time information. In the learning model, users react dynamically to changes in travel costs that can occur within-day and day-to-day time-frame. The application of congestion pricing to a simplified and realistic urban road network illustrates the importance of modeling users' behavior's dynamic and stochastic components to accommodate the efficient dissemination of congestion pricing strategies.

An EGT-based potential game theory approach is presented in Xiaojun and Haijun (2010) for studying the evolution of traffic assignments based on multiple criteria and multiple classes of users. The proposed model shows an admissible dynamic: an evolutionary process in which current utility evolves through myopic switching paths between fixed demand and elastic demand scenarios. In the case of strictly monotonic, continuous, and differentiable link travel time and inverse demand functions, the proposed model shows that the corresponding traffic assignment is the Nash equilibrium of a potential game. Moreover, a variable congestion pricing scheme is developed, allowing for a multiclass multicriteria traffic assignment to be determined by the system's current state.

Using an evolutionary game approach to study drivers, (Zhang et al., 2015) establishes two evolutionary stable strategies under various cost

conditions by establishing a game payoff matrix. This matrix is capable of verifying the existence of equilibrium in the toll channel selection game for drivers. Additionally, specific measures have been proposed to balance traffic flow distribution at the toll plazas. An equilibrium of the game is achieved by mixed strategy, according to the study; it is beneficial to refine the functional area of the toll plaza, arrange a reasonable amount of charging channels, and distribute the vehicles in weaving sections rationally to induce drivers to make strategic decisions close to equilibrium, which is an effective means of reducing congestion at toll

Based on the travelers' daily route adjustment behavior and the heterogeneity in their customary route adjustments, (Tan et al., 2015) investigates how congestion pricing can be implemented to minimize system cost and time. The model captures the heterogeneity of travelers through value-of-times. Firstly, this paper proposes a multi-class flow dynamical system that can reflect the behavior of travelers in adjusting their route preferences during traffic congestion in a toll-based transportation network composed of users from different classes of the population. After that, equilibrium properties and stability conditions are discussed. In addition, the authors develop a trajectory control model that minimizes the system's cost and time using dynamic congestion pricing. This method updates the link tolls periodically and forces the system to reach optimum levels in terms of system cost and/or time. Both analytical and numerical examples are used to demonstrate the method's efficiency.

An EGT-based dynamic congestion pricing scheme is proposed and implemented by the author (Ma et al., 2018) on a realistic urban highway in Athens, Greece, augmented with an evolutionary learning model. According to the model, several significant behavioral characteristics associated with the behavior of users in response to the congestion pricing strategy are identified. As an example of such characteristics, drivers' route choice and departure time choice behavior are subject to within-day as well as day-to-day adaptation mechanisms based on changes in travel costs, and there are heterogeneous classes of road users. The obtained results provide a foundation for understanding how dynamic congestion pricing schemes in urban highways can be deployed efficiently by modeling the stochastic learning processes of users. The findings of this study provide an opportunity for system operators and users to evaluate the acceptability of alternative congestion pricing schemes.

4.2.2. Dynamic congestion pricing: benefits to transits

In addition to improving public transportation speed and reliability, congestion pricing enhances public transportation services by lowering the costs of public transportation providers and strengthening the revenue sources available to public transportation providers.

Commuters who opt for public transportation instead of driving automobiles will benefit from road pricing since it absorbs commuters moving from car to bus or rail travel. Congestion pricing can improve the speed and productivity of existing express bus services on major routes by replacing congested traffic with free-flowing conditions so that these services are more appealing to commuters while lowering operating costs simultaneously. By reducing congestion, new bus rapid transit operations can be rapidly deployed in significant corridors without needing to invest in a large number of new vehicles or passenger facilities, which would result in the efficient deployment of innovative, high-performance bus rapid transit operations. With congestion pricing and limited capital investment, it is possible to improve the performance and variety of public transportation during peak periods. With the imposition of congestion pricing, public transit will be able to carry more passengers during rush hours, reducing the amount of time commuters spend in their cars during peak periods and can accommodate the shifts in commuter demand that result from congestion pricing.

Recent experience suggests that congestion pricing makes a difference to public transportation depending on the type of pricing strategy adopted. In the current market conditions, variably priced lanes have been found to have a limited effect on public transportation usage,

whereas zones-based (cordons or areas) pricing has had a significant impact. Table 6 summarizes the benefits of few real-world studies of congestion pricing.

5. EGT-based traffic route guidance

Route guidance systems play an important role in traffic routing. In traffic routing, vehicles are redirected or rerouted in a traffic network according to changes in traffic conditions to improve traffic flow. Several objectives are pursued during the routing process (Chen et al., 2012). Researchers focused most of their attention on finding the shortest distance when developing routing technology in the early years. Although traffic networks and routing processes have evolved, it is acceptable to sacrifice some distance over time to avoid traffic breakdowns. When dealing with congestion problems, it is most important to normalize traffic flow, regardless of time and distance considerations (Varga, 2022).

The first step in all prior routing methods is to construct a model of the road network based on the current status of the road network or the predicted traffic flow (A. Paricio & Lopez-Carmona, 2019; A. Paricio & Lopez-Carmona, 2019; Pi and (Sean) Qian, 2017). Next, algorithms such as the traditional Dijkstra algorithm are applied in order to determine the optimal or approximate optimal route for an origin and destination pair (Wang et al., 2017). The route is, however, optimal for a single vehicle. Managing a large number of vehicles requires coordination between vehicles (Varga, 2022). An urban centralized traffic controller coordinates all navigating vehicles by suggesting routes (Dündar et al., 2021; Sluijk et al., 2022; Asghari & Mirzapour, 2021). In the presence of a large number of vehicles, centralized approaches may suffer from scalability issues due to the fact that they are responsible for an entire region. To address the scalability issue, it is possible to include computationally inexpensive metrics for rerouting vehicles, increasing the ease of handling large volumes of traffic by the central controller. The decentralized approach is another option to explore, as vehicles would not rely on a centralized system to decide which route to take. However, this approach has some drawbacks, including the fact that it is often insufficiently informed about network conditions or requires an effective communication network to work. Accordingly, Manolis et al. (Pillac et al., 2013) conducted a comparative study of centralized and decentralized approaches for traffic light control.

ITSs facilitate real-time information sharing between transportation system operators and road users (i.e., vehicles) by integrating communications technology into vehicles and infrastructures. As a component of intelligent transportation systems, vehicular ad-hoc networks (VANETs) are widely considered important, and they can significantly contribute to intelligent decision-making by facilitating Vehicle to Everything (V2X) consisting of vehicle-to-infrastructure (V2I) communication, vehicle-to-vehicle (V2V) communication, vehicle-to-network (V2N) communication, and vehicle-to-pedestrian (V2P) communication (Hussain et al., 2021). Vehicular Networks have specific characteristics that make challenging decision-making processes for various intelligent devices, such as vehicles and roadside units (RSUs). These characteristics include the mobility of vehicles, the instability of the network topology, and the low bandwidth. Consequently, trade-off decisions may be required between goals contrary to or in conflict, such as throughput and delay (Kandasamy & Mangai, 2021).

In VANET, there have been some applications using traditional game theory; however, EGT's use is growing for the following reasons. An evolutionary equilibrium (i.e., the population's chosen strategies won't change over time) is an elegant solution that ensures stability. In evolutionary games, the strategy is changed slowly to reach the solution over time, while in traditional games, the decision is made immediately. A replicator dynamics analysis can provide insight into the trajectory of players' strategies (Wu et al., 2013).

Table 6

Real world case studies and benefits of congestion pricing.

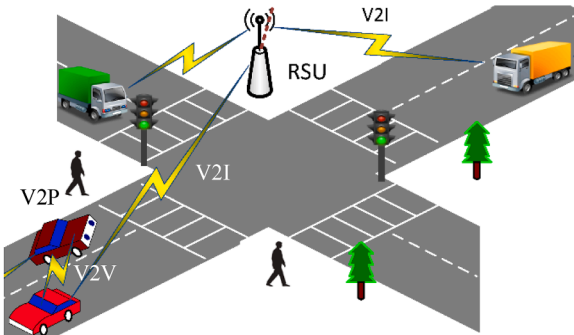
	Inception	Type of schemes	Charge area	Supplementary measures	Congestion Reduction Traffic volume	Others	Emission reduction
Singapore (Phang & Toh, 2004)	1998	Cordon scheme (mainly)	Three cordons in the CBD and expressways	Parking fees, car ownership taxes, park and-ride areas.	–44% car trips (ALS) –15% car trips (ERP)	travel speed increased 18–22 to 24–28 MPH, bus ridership +15%	–10% to - 15% CO ₂ and other GHG emission
London (Croc, 2016)	2003	Zonal schemes	21 km ² area in city center, extended to west	Promoting bus services	–33% car trips	–30% traffic delays, +38% bus ridership	–16% CO ₂ , –15.5%PM10, –13.5% NO
Stockholm (Börjesson et al., 2017)	Trial: 2006 Permanent: 2007	Cordon scheme	Cordon around 30 km ² inner city area	New bus lines, mass transit service improvement, park and-ride facilities	–22% across cordon, –16% within cordon	–30%to - 50% traffic delays, +4%- 5% transit system usage	–14% CO ₂ , –9% PM10, –7% NO _x
Milan (Beria, 2016)	Trial: 2008 - 2011. Permanent: 2012	Cordon scheme	8 km ² area in city center	New bus lanes, decrease on street parking	–14% car trips (Ecopass) –30% car trips (Area C)	+4% travel speed	–18% PM10, –10% CO ₂
Gothenburg (Kristofferson & Börjesson, 2018)	Trial: 2013 Permanent: 2014	Cordon scheme	A circle cordon with two expressways.	New bus lanes, parking restrictions.	–12% car trips	few% travel time savings	has not been observed

5.1. EGT perspective of vehicle routing game

When dealing with vehicle routing problems at intersections, the focus is primarily on the number of traffic flows that should be diverted to each link. Each vehicle makes its own decisions as it competes for the same traffic resource. The scenario can therefore be considered a game process (Zhang et al., 2016). An illustration of the urban traffic framework can be found in Fig. 11. An advanced driving assistance system is equipped with every vehicle in this road network, capable of communicating with RSUs, which can be categorized as standard and game RSUs. A standard RSU is primarily responsible for collecting and publishing traffic information, while a game RSU is located at intersection S and manages the game (Wu et al., 2013).

In population games (Fox & Shamma, 2013), simultaneous interactions can be modeled by observing the following three properties. A first feature is a large number of agents capable of making independent decisions. Secondly, the choices of a single agent do not significantly affect the outcomes of others. Lastly, an agent's payoff depends only on the distribution of its own strategy and the strategies of its neighbors. Consequently, the vehicle-vehicle game can be modeled as a population game, defined by $G = SP$, where P and S represent the set of populations $p \in P$ (a group of vehicles is represented as a population p) and the set of strategies available to vehicles. A vehicle trying to cross through an intersection with game RSU is considered one of the agents in the population. A number of categories of vehicles can be considered for a more realistic scenario. Based on (Lu et al., 2019), we have constructed a generic game for traffic routing.

It is commonly accepted that actions taken at an intersection, including turns to the left, turning to the right, turning to the back, and going straight, are considered pure strategies. In a population p , A_p

**Fig. 11.** An illustration of the urban traffic framework with vehicular network.

represents a set of agents, while $S_p = s_{k,p}$ ($k \in 1$ to n) represents a set of pure strategies. Each agent in the population adopts a pure strategy, and the population state is $X_p = x_p^k$ ($k \in 1$ to n), where x_p^k indicates the percentage of agents that select strategy $s_{k,p}$. We are seeking a social state $\Delta X = \{X_p\}$ where no vehicle could find a better path than the one it is currently taking when society P is under this state.

One can evaluate the payoff of each strategy based on the travel time to the intermediate destination. Assume that $c_{k,p}$ represents the payoff of strategy $s_{k,p}$ and $\bar{c}_p$ represents the average payoff of population p . Further, the $d_{cross,k,p}$ and $d_{link,k,p}$ represent delays associated with crossing intersections and travel times associated with running along links in strategy $s_{k,p}$, respectively. In addition, to indicate whether strategy $s_{k,p}$ passes through the link l_j , a binary number $R_{l_j}^{k,p}$ is introduced. Therefore, the detailed explanation of the payoff function and its constraints are explained in Eq. (6)–(7). The BPR formulation given in Eq. (8)–(9) is used to evaluate the delay of running on links.

$$\bar{c}_p = \sum_{s_k \in S_p} x_p^k \cdot c_{k,p}, \quad \sum_{k=1}^n x_p^k = 1, \quad \text{and } 0 \leq x_p^k \leq 1 \quad (6)$$

$$c_{k,p} = -(d_{cross,k,p} + d_{link,k,p}), \quad \text{where, } d_{link,k,p} = \sum_{l_j \in L} R_{l_j}^{k,p} \cdot d_{link}^{i,j} \quad (7)$$

$$d_{link}^{i,j} = d_{free}^{i,j} \left(1 + \alpha \left(\frac{n_j^i}{cap_j^i} \right)^\beta \right) \quad (8)$$

$$n_j^i = \sum_{p \in P} \sum_{s_{k,p} \in S_p} x_p^k \cdot |V_p| \cdot R_{l_j}^{k,p} + n_0^i \quad (9)$$

Where, on the link l_j , the free-flow travel time is denoted as $d_{free}^{i,j}$, n_j^i and $n_{j,0}^i$ are representing the total and the current volume of vehicles attempting to travel on the link l_j , while the number of agents participating in the game is denoted as $|A_p|$. α and β are constant parameters. For each population p , the population state (X_p) can be determined as $X_p = x_p^k$ ($k \in 1$ to n). No better strategy can be adopted under this population state which each agent is currently pursuing. Alternatively, we seek to determine Nash equilibrium for the population game as mentioned in Eq. (10).

$$NE(G) = \left\{ \Delta(x) : c_p^i \geq c_{k,p}, \quad \forall s_{k,p} \in S_p - \{s_p^i\}, \forall a_p^i \in A_p \right\} \quad (10)$$

The main challenge associated with the game is that although all population games contain at least one Nash equilibrium, the payoff

matrix and number of populations are large for this game. This leads to a high level of complexity in computing Nash equilibrium.

Since it is difficult to find the Nash equilibrium for the described population game G , the first step is to convert it into a dynamic model using a revision protocol. Under the conditions of a social state, the revision protocol describes the likelihood of an agent switching from strategy $s_{i,p}$ to $s_{j,p} \in S_p - \{s_{i,p}\}$. Evolution leads to the convergence of dynamics at some point as it evolves.

According to EGT, replicator dynamics indicate the rate at which strategies are switched. The replicator dynamic is able to achieve rest at the Nash equilibrium for a population game and each one of the pure states for the population game. Generally, a pure state is characterized by agents choosing the same strategy within a population, which means that over a large number of agents, that strategy is chosen as a percentage, with other strategies as a percentage of 0). A replicator is a classic method of imitating revisions. As a result, no interior solution trajectories converge to non-Nash rest points for an imitation dynamic of the population game (Lopez Mejia et al., 2022). In this case, the dynamic will converge to a Nash equilibrium if the initial state does not represent a pure state. A replicator is defined in Eq. (11). Based on this concept, the researchers have proposed various efficient traffic routing algorithms with replicators.

$$\frac{dx_p^k}{dt} = (c_{k,p} - \bar{c}_p)x_p^k \quad (11)$$

5.2. Review of existing literature on EGT based vehicle routing

The research community has made significant progress in traffic routing using EGT by developing both centralized and decentralized methodologies. In order to achieve a balanced usage of a road network, (Galib & Moser, 2011) has developed an EGT-based minority game model that assists drivers in making independent decisions. Neither real-time information nor communication between drivers is required in the proposed model; only past experience on driver route choices is used. The experimental results show that near-optimal traffic distribution in a network can be achieved using the EGT approach. The study in Elhenawy et al. (2015) claims that a cooperative adaptive cruise control system at uncontrolled intersections could be controlled through an EGT-based chicken-game algorithm using cooperative adaptive cruise control. Using the proposed algorithm, existing intersection control systems such as traffic signals, street signs, and symbols can be replaced with the robotic capabilities of autonomous vehicles in the future. According to simulation results, there was a 49 percent average reduction in vehicle travel time and an 89 percent reduction in vehicle delay compared to an all-way stop sign control.

Authors in Lu et al. (2019) have modeled the vehicle routing problem as a population game. An algorithm based on EGT, which can be distributed cooperatively to achieve the Nash equilibrium, is proposed. Using a simulation experiment, a comparison is made between the proposed algorithm and five other algorithms to evaluate its performance. According to the experiment results, the proposed algorithm can effectively divert traffic at the intersection. A distributed vehicle routing framework based on edge-assisted distributed agents developed in Zhu et al. (2019) facilitates real-time routing decisions by interacting with other agents i.e., V2V interaction. To find the Nash equilibrium, we propose a scalable multi-agent learning algorithm based on the model of agent interaction as a population game. The proposed multiagent learning algorithm offers a stage learning algorithm that incorporates best-response dynamics while taking into account the bounded rationality of each agent. The findings demonstrate efficient convergence of Nash equilibriums, which reduces the average travel time of vehicles at intersections and maximizes throughput. An incentive framework using decision trees and EGT is presented in Mei et al. (2017) for routing vehicles and creating smart incentives based on travelers' travel patterns. This information includes travelers' mobile patterns, preferred modes of

transportation, and the traffic volume on their routes. The proposed framework was validated by a series of MATLAB simulations, demonstrating that it may be an effective method for incentivizing travelers to change their travel routes and modes. (Shittaka & Nagae, 2017) describes a novel traffic routing concept that uses EGT and randomized algorithms to ensure traffic movements' reliability throughout a road network. A moderately sized network with several dozens of routes was demonstrated to be capable of estimating the travel time reliability using the proposed method. The model, however, requires a high calculation load for more extensive networks due to the state space's dimensionality in the network's day-to-day dynamic.

An edge computing-based hierarchical traffic-balancing routing method for connected vehicles has been proposed in Han et al. (2020). This system divides the road network into grids and guides traffic flows between grids using an improved back-pressure algorithm. Further, the vehicle routing at each intersection is coordinated by the EGT game method in order to maintain a balanced intra-grid traffic flow. According to simulation results, the algorithm increases throughput while reducing travel time and balancing road network resources. The authors have explained in Lin et al. (2018) how a social vehicle route selection algorithm can be used for selecting the optimal route for vehicles. Optimal vehicle routing is achieved by using social clustering and EGT to improve traffic control performance. To ensure a stable survival strategy and maintain dynamic equilibrium, the EGT attempts to find a suitable strategy. In terms of clustering the vehicles and reducing traffic congestion, the simulation results indicate that the algorithm adopted is highly effective. Future research can consider the driver's personal characteristics to improve traffic control. To enhance the development of online data processing algorithms, researchers can also develop artificial intelligence algorithms considering the background of massive data.

The articles discussed above have focused on traffic routing, which includes two parties: traffic routing operators and vehicle drivers. However, realistic scenarios often involve multiple objectives problems with more than two players involved. With these considerations in mind, (Chen et al., 2020) proposes a distributed coordinate routing algorithm with light timing algorithms supported by edge computing architectures so that it can optimize both traffic routing and traffic signal timing simultaneously. Using the EGT, this algorithm achieves Nash equilibrium. The effectiveness of this joint algorithm has been demonstrated through experimental testing, mainly when dealing with non-uniform traffic flow situations. Further research may investigate how the distributed controllers can be deployed at more intersections and synchronized simultaneously. Furthermore, (Su et al., 2007) has constructed an EGT-based game model for simulating the evolution of urban public transportation networks, which consists of three players: government agencies, companies that provide urban transportation services, and motorists. According to the authors, network manipulators produce better results in some systems than nodes performing games between them.

As mobile intelligent objects in smart cities, vehicles collect, store, compute, and communicate data that ITS uses. For autonomous vehicles to be routed efficiently, data from the Internet of Vehicles (IoV) must be collected and analyzed efficiently. For efficient autonomous vehicle routing, the authors in Zhao et al. (2019) propose implementing big data acquisition and analysis in IC-IoV using an edge-MapReduce architecture. By analyzing traffic data and user needs, the IC-IoV provides traffic tasks and driving routes corresponding to those tasks. In addition, the authors have realized that the EGT is an effective incentive mechanism for allocating tasks and sharing information. Simulation results indicated that the proposed strategy encourages data sharing among connected vehicles and promotes information sharing in IoV. Future research on developing intelligent networks with capabilities for big data analysis may be considered to provide a more comprehensive range of services to the Internet of IoV. A research direction for the future is also to protect edge information in consideration of user privacy and

traffic safety.

Vehicles accessing RSUs (V2I) face two challenges; (1) Due to the high mobility of vehicles, it is difficult to maintain an end-to-end connection between them. In the case of RSUs with limited bandwidth, this will result in disorderly competition behavior between vehicles, which will lead to multiple RSUs having overlapped areas, at which time access to RSUs will be crucial for increasing vehicle throughput. For the problem of RSU access in VANET, (Wu et al., 2013) has established a game model with bounded rationality based on the EGT paradigm to formulate competition among vehicles within and among different communities. The authors have examined how the dynamics of ESS evolved as vehicles pass through RSUs. Vehicles access RSUs depending on their payoff, which depends on the throughput and cost of the RSU. Using the replicator dynamics, authors have mathematically modeled the dynamics of access to RSUs over time, which describes the proportion of vehicles accessing different RSUs. The authors used a realistic vehicular traffic model to simulate the evolution of the game, as well as the effect of the ESS on its performance.

In Zhang et al. (2014), the authors have provided an EGT framework to analyze the relationship between bandwidth occupation and information dissemination in the case of vehicular networks. Authors used EGT to guide the behavior of each member in VANET to achieve satisfactory packet forwarding performance. The significant discovery is as follows: a) The cooperation ratio in convergence for EGT is proportional to the synergy factor. b) An EGT-based packet forwarding scheme can achieve a similar information delivery ratio while reserving more bandwidth.

In Zhang et al. (2016), the authors investigated how vehicles cooperated in the VANET to disseminate data. To reduce bandwidth consumption and improve data dissemination rates, it has been proposed that nodes periodically modify their strategy (cooperation/defection) in response to EGT. The simulation results indicate that various information dissemination schemes can be effectively traded off for different network environments by carefully controlling the number of cooperators via EGT. The authors of this study have investigated the distribution of cooperators under the proposed scheme. Based on the

results, the following conclusions were reached. 1) There is always a non-uniform distribution of cooperators. 2) A network's mean node degree and how much data needs to be disseminated determine whether high-degree or low-degree nodes have more incentive to cooperate. 3) Whenever unnecessary broadcasts are pre-canceled by overhearing duplicate neighboring transmissions before they are transmitted, the high-degree nodes have a greater incentive to cooperate. The researcher may consider a future study of the evolution game based on mixed strategies as a possible avenue for future research. Further, to enhance the throughput of hybrid V2V communications, (Ming et al., 2022) proposes an evolutionary game-based algorithm based on strategy selection. The first step is to design multiple V2V hybrid communication strategies that can use different communication modes for different kinds of packet transmission, in order to ensure flexibility of communication. An evolutionary game is then used to model the selection of hybrid V2V communication strategies, and an algorithm is proposed for selecting the best strategy. Based on the locally estimated payoffs for various V2V communication strategies and the average payoff broadcast by the base station, vehicles are able to adjust their V2V communication strategies independently in the proposed algorithm and could ultimately evolve into satisfactory and stable V2V communication strategies. According to simulations, the proposed algorithm is robust to the invasion of a few vehicles using the mutated strategies, even when there are slight errors in payoff estimation. The algorithm achieves better performance than conventional algorithms. A comparative analysis of the above literature is summarized in Table 7.

5.3. Route guidance and driver's response

Route guidance system is one of the major components of intelligent transportation networks. An optimal route from the driver's current location to the destination is calculated using route guidance systems and is updated based on events such as traffic jams, incidents, bad weather conditions, etc. the route guidance system is well classified in Fig. 12. Various tools have been explored to study the different aspect of route guidance system, such as rule-based reasoning (Guo et al., 2006),

Table 7
A comparative analysis of the EGT based literature under traffic routing.

Ref	Developed technology	Communication	Players	Objectives	Results
(Galib & Moser, 2011)	EGT-based minority game	No communication	Vehicles and TSO	Single	Optimal traffic distribution
(Elhenawy et al., 2015)	EGT-based chicken-game	No communication	Vehicles and TSO	Single	Reduction in vehicle travel time and vehicle delay
(Lu et al., 2019)	EGT-based population game	Vehicular network	Vehicle and RSU	Single	Effective traffic diversion
(Zhu et al., 2019)	EGT-based population game and MAS	Vehicular network	V2V interaction	Single	Reduces average travel time at intersections & maximizes throughput
(Mei et al., 2017)	EGT-based incentive framework	Vehicular network	Vehicles and TSO	Single	Optimal traffic distribution
(Shittaka & Nagae, 2017)	EGT and Markov chain Monte Carlo	Not mentioned	Vehicles and TSO	Single	Ensure traffic movements' reliability
(Han et al., 2020)	EGT, edge computing, and improved back-pressure algorithm	Vehicular network	Vehicles and RSU	Single	Reducing travel time and balancing road network resources
(Lin et al., 2018)	Social IoV clustering and EGT	Vehicular network	Vehicles and RSU	Single	Reducing traffic congestion
(Chen et al., 2020)	EGT and edge computing	Vehicular network	Vehicle, TSO, TSC	Multi	Traffic routing and TSC timing
(Su et al., 2007)	EGT	No communication	Government, TC, and motorists	Multi	Each manipulator enhances their benefit
(Zhao et al., 2019)	EGT, IC-IoV, and edge-MapReduce architecture	Vehicular network	Vehicle and RSU	Multi	Encourages data sharing and promotes information sharing.
(Wu et al., 2013)	EGT	VANET	Vehicle and RSU	Single	Efficient information dissemination among the vehicles and communities
(Zhang et al., 2016)	EGT	VANET	Vehicle and RSU	Multi	Reducing bandwidth consumption and improving data dissemination rates
(Zhang et al., 2014)	EGT	VANET	Vehicle and RSU	Multi	Efficiently bandwidth occupation and information dissemination
(Ming et al., 2022)	EGT and LTE-V2X	VANET	V2V interaction	Multi	Reducing bandwidth consumption and autonomous data exchange among vehicles

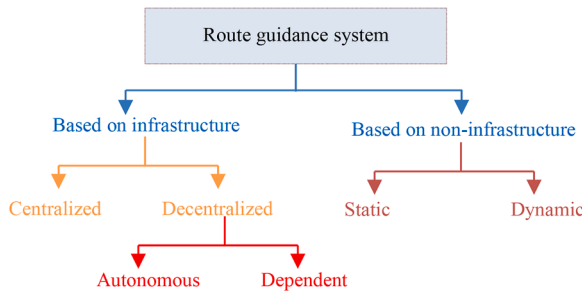


Fig. 12. Generic classification of route guidance system.

cellular automata (Chen et al., 2021), VANETs (Sakoguchi et al., 2019), fuzzy -neural networks (Mi & Liu, 2017), genetic algorithms (ZOU et al., 2007), neural networks (Lin & Liu, 2014), agents (Namoun et al., 2021; Aliali & Benchaiba, 2017), etc.

This review article is focused on the EGT, thus, only those that comes under the scope of EGT is covered. In this line, the variable message signs (VMS) function as a critical component of a transportation system that aims to make it as efficient and effective as possible, specifically in terms of optimal route guidance. The VMS provides travelers with the following information: suggestions regarding alternative routes, incident notifications, congestion conditions, special event announcements, etc. Several studies have demonstrated that VMS displaying route guidance information can influence drivers' route diversion decisions, leading to better system performance. It is worth noting, however, that the percentage of drivers who make a change of route choice may vary significantly, with (Erke et al., 2007) estimating that between 5% and 25% of drivers make such a choice. According to Cummings (1994), only 4–7% of all drivers follow any recommendations. Road blocking induced 30% more drivers to divert their vehicles (Ramsay & Luk, 1997). (Chatterjee et al., 2002) reports that only a third of all drivers notice VMS regarding network difficulties, and few get redirected from their regular routes.

VMS information and driver responses can be elicited using Stated Preference and Revealed Preference. Revealed Preference analysis analyses driver behavior under real-world situations based on driver reports and observations of actual driver behavior in field studies. In the Stated Preference approach, drivers are presented with hypothetical travel alternatives in order to analyze their behavior. This method has been used in analyzing modes of transportation, and it is now being used more frequently to analyze drivers' route choices under the influence of real-time traffic information Peeta & Ramos (2006). Studies have examined drivers' revealed and stated preference responses to VMS information. An analysis of the VMS based on driver questionnaires, traffic network modeling, and network monitoring is provided in Majumder et al. (2013). (Peeta & Ramos, 2006) evaluated drivers' responses by providing them with information regarding road conditions with and without incident conditions. According to AlKheder et al. (2019), Kuwaiti drivers' responses to VMS information are most strongly impacted by the location of the incident and the message content. However, these studies have neglected the dynamic drivers' behavior.

Driver's response to the information plays a significant role in the success of VMS strategies. In response to the VMS information, the driver may opt options available to them: accepting the recommendation or refusing it. Therefore, it is imperative to understand how drivers make decisions based on real-time information. The decision-making problems best fit into game theory.

To understand the driver's response from the GT perspective, let's assume a simple route choice problem illustrated in Fig. 13 consisting of two routes R_1 and R_1 between intersections, O and D , explained in Li and Wang (2011).

Point C represents the location of VMS. When drivers are driving on route R_0 , the VMS recommends an alternative route to them. Drivers

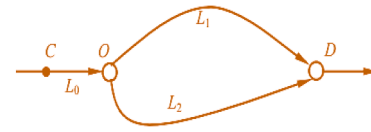


Fig. 13. A generic two-route model.

compare their experience and knowledge before making a decision. In addition to acting on VMS recommendations, the driver can refuse the recommendation. A game theory model can be used to describe this decision-making process. Which can be represented as $G = \{A, S, U\}$. The term A refers to a set of players, i.e., drivers. A set of strategies is defined by $S = \{s_1, s_2\}$, s_1 and s_2 denote the decisions to accept or reject the recommendations made by the VMS. A player's payoff is defined as U , generally referred to as travel time. Shorter travel time refers to a higher payoff and vice versa. The constructed payoff matrix can be represented as Eq. (12).

$$U = \begin{matrix} & s_1 & s_2 \\ \begin{matrix} s_1 \\ s_2 \end{matrix} & \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} \end{matrix} \quad (12)$$

When a player's strategy is s_1 and the opponent's strategy is also s_1 , u_{11} represents the payoff that the player s_1 receives. Strategy s_1 players receive u_{12} as a reward when they play against strategy s_2 players. When a strategy s_2 a player plays against a strategy s_1 player, u_{21} represents the payoff of the strategy s_2 player receives. When a strategy s_2 a player plays against another strategy s_2 player, u_{22} is the payoff he receives.

Based on the above concept, (Choi & Jang, 2004) has developed a more systematic approach to VMS operation. Road users and operators are modeled as players, along with payoff matrices representing user equilibrium and the system's optimal status. The authors in Lee and Shin (2011) have explained how the VMS system operator can select the drivers' route based on one of four electric messages sent from the VMS system to the drivers through a framework of the cheap-talk game. However, these studies are based on classical GT and have multiple limitations (referred to Section 2.2.2). Thus, dynamic decision-making models using EGT are recommended.

5.3.1. EGT perspective of drivers' response

In response to the VMS recommendation, several drivers selected strategy s_1 , whereas others selected strategy s_2 . All drivers have three characteristics: diversity, selection, and replication. Since the selection process is based on the payoff, an approach that provides a higher return is preferred to a relatively lower one. The replication preserves replicate or inherit the drivers' properties from one choice to another.

Drivers can choose from various strategies in response to the VMS recommendation and receive the corresponding payoffs. Each time a decision is made, the drivers can assess the payoff and adjust their strategy selection accordingly. Different strategies can lead to different payoffs for drivers as they evolve. As a result, strategies selected by drivers would change in accordance with the principle of survival of the fittest. In the event of increasing payoffs, the drivers will adhere to their strategy, and in the event of declining payoffs, they will change their strategy. Thus, the proportion of people who choose s_1 changes from one choice to another or changes over time. EGT can be used to study how drivers' frequency of strategies changes.

The evolution of a system may be viewed as continuous in terms of the time it takes for a choice to be made. At the k^{th} decision, let's $y(k)$ represents the proportion of individuals who adopt strategy s_1 . This means the proportion of individuals who adopt the strategy s_2 is equal to $1 - y(k)$. The differential equation in Eq. (13) describes how the population evolves according to the EGT (Sumaila & Apaloo, 2002). Based on the above equation, one can determine how fast a population's percentage of people using strategy s_1 will change in the future. Eq. (13)

describes the replicator dynamics.

$$\dot{y} = \frac{dy}{dk} y(U_{s_1} - \bar{U}) = y(1-y)(U_{s_1} - U_{s_2}) \quad (13)$$

Where, $U_{s_1} = yu_{11} + (1-y)u_{12}$ and $U_{s_2} = yu_{21} + (1-y)u_{22}$ denote the strategy's fitness of s_1 and s_2 while the average fitness is represented as $\bar{U} = yU_{s_1} + (1-y)U_{s_2}$. Thus, Eq. (13) can be modified as Eq. (14).

$$\dot{y} = y(1-y)[y(u_{11} - u_{21}) + (1-y)(u_{12} - u_{22})] \quad (14)$$

Eq. (14) represents an evolutionary model for selecting a driver's strategy under the guidance of VMS recommendations. And also, Eq. (14) describes the evolutionary process by which the proportion of drivers adopting strategy s_1 changes over the course of time. Now, let's suppose $f(y) = \dot{y}$, in that case, $f(y) = (1-2y)[(u_{12} - u_{22}) + y(u_{11} - u_{21} + u_{22} - u_{12})] + (y - y^2)(u_{11} - u_{21} + u_{22} - u_{12})$. According to the EGT, y^* is the evolutionary stable state if $f(y^*) = 0$ and $f'(y^*) < 0$ are simultaneously satisfied.

Based on the above model, the authors in Li and Wang (2011) proposed a decision-making model of strategy selection that examines how drivers respond to variable message signs' recommendations. The proposed model offers two strategies for drivers, either accepting or refusing the recommendation. An analysis of stable states and stable strategies is presented. The study results show that if and only if the drivers' payoff of accepting the recommendation is greater than the driver's payoff of refusing the recommendation over the course of time, then the proportion of drivers who accept the recommendation will converge to 100% over time. To implement VMS strategies, the payoff for accepting recommendations must be greater than the payoff for refusing the recommendations, and the difference between the two must be as substantial as possible. Meanwhile, when drivers use VMS recommendations for the first time, the system is expected to make a positive impression. The purpose of these is to support the decision-making process for VMS strategies. Also, (Kim et al., 2014) attempts to analyze a driver's response based on information obtained from the VMS traffic system employing EGT. It was found that in this study, drivers accepted information only when it offered the best reward, and if the ratio of accepting information is able to find a balance over time, it converges gradually into a stable evolutionary state as the behavior characteristics of drivers are applied to EGT. Consequently, it is difficult for a driver to predict the strategy of another driver. Through EGT, it is expected that VMS could provide strategic information when drivers repeat between groups, and reasonable judgment is based on experience.

6. Transport safety

Traffic accidents not only cause casualties but also seriously affect economic development and social stability. Various accident-avoidance strategies, including optimal interactions, driving strategies, speed recommendations, and accident/collision prediction, have been developed to prevent accidents and ensure transport safety. The interactional behavior of vehicles, pedestrians, and TSO at signalized as well as non-signalized intersections, including crosswalks, is well covered in the Section dealing with TSC.

6.1. EGT perspective of optimal driving strategy

EGT perspective, the concept of an optimal driving strategy can be examined by considering the dynamics of interactions between different drivers on the road. In this context, drivers can be seen as players in a game where their strategies determine their driving behaviors. EGT offers a framework to analyze how drivers make decisions and adjust their strategies based on interactions with other drivers. One aspect of an optimal driving strategy is to maximize individual benefits, such as reaching the destination efficiently and safely. However, in a game-theoretic perspective, an optimal strategy also takes into account the

behavior of other drivers and the potential for cooperation or competition on the road. In EGT, optimal strategies can be identified through the concept of equilibrium. Nash equilibrium, for instance, occurs when no individual driver can unilaterally improve their own outcome by changing their strategy. This equilibrium represents a stable state where drivers' strategies are aligned, and no driver has an incentive to deviate from their chosen course of action. Moreover, evolutionary dynamics can shed light on optimal driving strategies. Through repeated interactions and learning, drivers may adapt their strategies over time to achieve more favorable outcomes. Evolutionary algorithms, such as genetic algorithms or reinforcement learning, can be employed to model and analyze the evolution of optimal driving strategies in complex traffic scenarios. Additionally, cooperative strategies can also be explored from an EGT perspective. Cooperative game theory investigates how drivers can form coalitions or cooperate to achieve mutually beneficial outcomes. For example, cooperative lane-changing strategies or cooperative merging behaviors can optimize traffic flow and reduce congestion.

In literature multiple EGT models for optimal driving strategy have been developed, especially considering rear-end collisions. A rear-end collision is a commonly observed incident in densely populated urban traffic, where driving strategies significantly contribute to these accidents. Liability rules serve as a primary mechanism that determines how the costs associated with a crash are allocated between drivers and their respective insurance companies, depending on the circumstances involved. In Chatterjee and Davis (2013), an exploratory study was proposed, utilizing EGT to assess the effects of liability rules on rear-end crashes. The driving behavior of a driver in a typical car-following scenario between two vehicles was measured, and it was correlated with a utility representing the expected outcome based on the driver's actions and the actions of their opponent. To model this interaction, a two-player game was employed, simulating an evolutionary process that involves random and repeated pairings of drivers with various driving behaviors. The games' outcomes determined specific driving strategies followed by a certain percentage of drivers in the subsequent phases. The stable long-term distribution of driving strategies was utilized to predict the probability of a rear-end accident. Surprisingly, it was found that a population in which all drivers aim to avoid crashes does not exhibit evolutionary stability when the direct crash costs are equally distributed among them.

To further investigate the interactions among vehicles with various driving strategies and their cooperation and competition, the authors of (Jiang et al., 2021) have developed a coevolutionary algorithm combining the multi-objective optimization algorithm with the incentive mechanism of the survival of the fittest reproduction law. The proposed system structure has achieved a stable equilibrium state, and the optimal strategy for group driving has been developed. Among six typical strategies, the rational strategy performed the best in the Luxembourg SUMO Traffic Scenario simulation with 40,000 vehicles. At the same time, there has been a 72% reduction in accident rates while an overall 30% increase in average speeds. Developing an optimal driving strategy with automotive vehicles is enlightening based on the results of multi-vehicle and multi-objective coevolution.

In addition, through an analysis of how time and safety factors impact the behavioral strategies of vehicles on the road, a network game evolution model between drivers was developed by Lv et al. (2022). This model takes into account the behavior strategies of individual vehicles as well as their neighboring vehicles and investigates the competitive relationship between different vehicle types. The research findings indicate that the selection of behavior strategies, such as speeding and lane changing, is influenced by the expected benefits influenced by factors such as time, safety, and the strategies of neighboring vehicles. Moreover, the critical value of expected benefits is determined in this study. Unethical driving strategies such as illegally occupying emergency lanes raise safety concerns. Thus, (Feng et al., 2017) investigates the evolution of drivers' and traffic administration departments' strategies for addressing illegal occupancy behavior at a microscopic level.

Initially, the factors influencing players' decision-making are examined, along with the impact of emotions, before developing a game model between drivers and TADs. Secondly, the local stability principle of a dynamic system is applied to analyze the ESS based on each of the different strategy payoff relationships. Based on the evolutionarily stable strategies, improved strategies are then suggested. Finally, a dynamic system based on survey data was also tested using MATLAB to verify the ESS determined by the local stability principle. After adjusting the values of the payoff influence variables, a comparison of evolutionary signs of strategic selection progress was also conducted between the two players.

6.2. EGT perspective of vehicle speed recommendation

The EGT perspective can be applied to the field of vehicle speed recommendation to optimize driving behaviors and improve overall transportation outcomes. In this context, EGT provides a framework for analyzing the strategic interactions among individual drivers and their impact on the system as a whole. The concept of vehicle speed recommendation aims to guide drivers towards optimal driving speeds that balance various objectives such as fuel efficiency, safety, and traffic flow. Traditional approaches often focus on specific factors like fuel consumption or travel time without considering the broader dynamics of the transportation system.

Considering speeding as an essential parameter, (Li & Zhang, 2010) has developed a model that allows incentives and penalties to be used when deciding whether to enforce speed limits, inspect traffic managers, or not enforce speed limits, to reduce the speeding behavior of drivers. Based on the local stability of the Jacobian matrix, possible evolutionary stable states are analyzed. In light of the results, government departments should recognize and reward traffic managers for their commitment to their duties and penalize them when they fail. The penalties for speeding drivers should also be based on the individual circumstances of each individual. As a result, speeding may be prohibited in certain situations, and traffic safety may be improved.

In order to enhance transportation safety, (Asadi et al., 2022) has introduced an innovative model for providing safety-conscious speed recommendations that enable individual drivers to prioritize their secondary driving goals, such as eco-driving and safety enhancement. The proposed model employs an evolutionary game mathematical framework to generate personalized speed profiles for each vehicle, taking into account the need for a balance between the competing objectives of road users. Notably, this system does not rely on advanced vehicle infrastructure or inter-vehicle communications, making it accessible and feasible for a wide range of vehicles on the road. The aim is to empower drivers to make informed decisions regarding their driving speed that align with their safety priorities, thereby contributing to improved overall transport safety.

By adopting an EGT perspective, researchers and policymakers can take into account the strategic behavior of drivers and how their decisions influence the overall system. EGT models consider drivers as players in a game who make choices based on their own interests and the anticipated actions of other drivers. Through repeated interactions and learning processes, a stable equilibrium or optimal strategy can emerge.

EGT-based vehicle speed recommendation models leverage the insights from game theory to design mechanisms that incentivize drivers to adopt safer and more efficient driving behaviors. These models incorporate factors such as safety awareness, eco-driving, and individual driving goals into the decision-making process. By considering the interactions between drivers and the trade-offs between different objectives, EGT-based approaches can provide personalized speed recommendations that align with the desired outcomes for both individual drivers and the transportation system as a whole.

The advantage of using EGT in vehicle speed recommendation is that it captures the complex dynamics of driver behavior and the interactions between drivers on the road. It enables the development of strategies

that promote safer driving practices, improve traffic flow, and reduce congestion. Furthermore, EGT-based approaches do not necessarily rely on advanced infrastructure or inter-vehicle communication systems, making them more accessible and practical for implementation.

6.3. EGT perspective of accident/collision prediction

The EGT perspective can be applied to the prediction and analysis of accidents or collisions in the field of transportation. By using evolutionary game theory as an analytical tool, researchers can study the strategic interactions among different actors involved in driving behaviors and examine how these interactions influence the likelihood of accidents. Traditionally, accident prediction models have focused on factors such as road conditions, vehicle characteristics, and driver behavior. While these factors are important, they often overlook the strategic decision-making of drivers and the influence of their actions on accident probabilities. By incorporating EGT into accident prediction, a more comprehensive understanding of the underlying dynamics can be achieved. In an EGT framework, drivers are viewed as players in a game who make decisions based on their own objectives and perceptions of other drivers' behaviors. The interactions between drivers, such as overtaking, lane changing, or maintaining a safe distance, can be modeled as strategic moves within the game. Through repeated interactions and learning processes, certain equilibrium strategies or patterns of behavior may emerge.

Prediction of accidents or collisions is an essential component of road safety since it prevents accidents before their actual occurrence. A scenario in Wang et al. (2021) in which intelligent vehicles perceive pedestrians crossing streets and predict a collision may occur is analyzed using evolutionary dynamics. The parameters of the conflict matrix between pedestrians and intelligent vehicles are assigned values based on factors such as the vehicle type and the radical degree of pedestrian behavior. A strategy is developed to determine whether an intelligent vehicle will pass or not based on the evolutionary paths they have followed and whether pedestrians will pass. This study showed that the evolutionary process could be used to make decisions regarding the behavior of intelligent vehicles, thereby preventing collisions with pedestrians crossing the street.

In Dai et al. (2021), authors developed an evolutionary game-based distributed hierarchical model predictive control algorithm for multi-agent systems in obstacle-filled environments. A strategic-tactical structure is designed, with the leader vehicle using predictive control for optimal control input while considering obstacle avoidance. Follower vehicles utilize distributed density-dependent Smith dynamics and predict their state using evolutionary game theory. They make simultaneous decisions based on predicted positions and incorporate predicted neighbors' positions to avoid collisions. Numerical simulations validate the algorithm's effectiveness in obstacle avoidance and collision prevention.

Based on replicator-mutator dynamics and complex Laplacian, (Guzman-Hernandez & Mojica-Nava, 2016) propose a leader-follower formation control algorithm. It consists of a networked multi-agent system that comprises several autonomous vehicles that keep a formation in a given area. For leader-follower formations, replication-mutation dynamics facilitate a faster convergence time and prevent potential collisions between followers. By modifying the strength of mutation and the frequency of the strategy, the algorithm can modify the consensus agreement in geometric formations while avoiding collisions between vehicles. The developed control algorithm is capable of maintaining a geometric formation in any path within the complex plane without requiring any modifications to the algorithm.

Moreover, (Wei et al., 2018) introduces a paradigm of evolutionary analysis to overcome vehicle-pedestrian collisions and establishes a dynamic model of how the conflict evolves between pedestrians and vehicles. The conflict between pedestrians and vehicles is represented by a game matrix that includes basic payoffs, conflict losses, waiting losses,

and mutual avoiding losses. It was found that conflict loss and waiting loss are of different sizes for pedestrians and vehicles, indicating that the system evolved in different directions. "The vehicle first," "the pedestrian first," "neither the vehicle nor the pedestrian goes first," and "both vehicles and pedestrians never yield to each other" are possible evolution directions. Furthermore, (Wei et al., 2018) defines opportunity loss as a concept in traffic and analyzes the sensitivity of the vehicle and pedestrian systems to mutual avoiding losses. A positive effect of mutually preventing the loss of pedestrians or vehicles is observed when it increases, whereas an adverse effect is observed when it is reduced. It must be noted that opportunity loss has the opposite effect of mutual avoidance loss, and it is a loss caused by the lack of opportunity. In Wei et al. (2018), the dynamic model has been established to provide a theoretical basis for controlling the macro-level conflict evolution between pedestrians and vehicles by developing a theory of conflict control. An excellent example of such a conflict can be seen when pedestrians and vehicles are contending for space in a city where "vehicles are always first."

From the above studies it is concluded that by studying the evolutionary dynamics of driver behavior, researchers can gain insights into how accidents or collisions may occur. The analysis focuses on understanding the incentives and trade-offs faced by drivers in their decision-making process. Factors such as risk-taking behavior, reaction times, and response to the actions of other drivers can be incorporated into the modeling framework. The EGT perspective allows researchers to identify critical factors that contribute to the occurrence of accidents and to analyze how changes in behavior, environmental conditions, or policy interventions can impact accident probabilities. By simulating different scenarios and observing the outcomes of the evolutionary game, it is possible to gain insights into the effectiveness of various strategies for accident prevention. Furthermore, the EGT perspective enables researchers to study the stability and optimality of driving behaviors within the context of accidents. By analyzing the Nash equilibria or other solution concepts in the game, researchers can identify strategies that promote safer driving and reduce the likelihood of accidents.

7. Challenges and future research directions

This section identifies two major gaps that can be addressed to improve existing knowledge. First, it is necessary to define the problems in light of travelers' design requirements and behavior more realistically; second, it is essential to develop more efficient solutions based on the advancement of computing technology and its limitations. Based on these two points, we have summarized the challenges and future work dealing with various aspects of the road transportation network.

7.1. Challenges with EGT

In EGT-based interactions, the outcome is conveniently encoded in a payoff matrix then coupled with a set of differential equations, called the replicator equations, determines how the population evolves. Replicator equations, however, have some limitations; (1) populations are perfectly mixed, meaning that offspring are placed at random locations and are not necessarily surrounded by their kind, (2) decisions are deterministic, i.e., there is no stochasticity involved, and (3) there is an infinite size of the population. The replicator equations predict the same outcome as the ESS in this limit. An actual population, however, cannot be infinite. It is also unlikely that they will ever be perfectly mixed.

The limitations listed above are merely the most obvious ones. Some further limitations of evolutionary game theory may be of greater importance than those of the mathematics involved in it. An excellent example is making decisions without always being deterministic, and the memory of past encounters can also influence decisions in some cases. A significant limitation of evolutionary game models is that all feasible strategies cannot be competed against once in an evolutionary scenario. Instead, the strategies that are competing are those that are

prevalent at present and those that are becoming more prevalent over time. Once-dominant strategy can be made extinct by a newcomer, while new strategies emerge to test their mettle against established ones. As a result, evaluating a strategy's success must consider the context to which the strategy is exposed, both in terms of time and space.

Additionally, as in classical game theory, reducing complexity in EGT (evolutionarily stable strategies) is a critical and challenging task. There are still many assumptions and limitations in the methods proposed to reduce complexity. First, a game with unique properties can be simplified by simplifying the payoff matrix, but it cannot be applied to general or complex real-life scenarios. This necessitates the development of more efficient algorithms to accommodate a nonlinear system into EGT-based models to generate an optimal or near-optimal solution, thereby reducing the complexity of the model.

7.2. EGT-based travel mode choice

Travel choice behavior is influenced by complex factors, but the relationship with transportation modes is still uncertain, posing challenges for correlation identification and accurate predictive modeling.

Previous research introduced the concept of the "indifference threshold," indicating that travelers are unable to distinguish between choices below the threshold. Instead, they opt for the choice with the highest overall value if it surpasses the threshold. Although this concept effectively explains travel behavior, there has been no investigation into the dynamic evolution of behavior concerning the indifference threshold. To address this gap, it is necessary to integrate logit dynamic evolutionary game models with the indifference threshold in order to analyze the development of travel behavior.

A dynamic approach provides insights into various travel mode choice models, including ridesharing, and enhances understanding of travelers' decision-making processes. These studies utilize traveler utility functions to assign parameter values. Model calibration can be partially achieved by estimating the percentage of travelers selecting car-sharing over private vehicles in a stable system using market share data. However, obtaining accurate market share data during system stability is challenging, particularly for emerging concepts like car-sharing. Future research should concentrate on improving the evaluation of travelers' utility functions, measuring psychological benefits, and realistically exploring and establishing parameters in numerical analysis.

7.3. EGT-based lane changing

Developed evolutionary game-based lane change (LC) models rely on two-player game models to generate decisions. However, these models have limitations. In complex urban traffic scenarios, where multiple vehicles change lanes simultaneously, the binary options of waiting or merging are unrealistic. Follower vehicles can also play a role by adjusting their speed to create merging opportunities.

In certain situations, sacrificing the interests of following vehicles may be necessary, such as when congestion requires lane changes to accommodate more vehicles. Current studies predominantly focus on two-player games, but this approach becomes ineffective during high-density traffic flows with simultaneous LC demands. While evolutionary game theory (EGT) has been extensively applied to highway LC models, its utilization in urban road scenarios is limited. Few studies have used game models to explain LC behaviors of buses or city electric vehicles, and there is a lack of game-theoretical analysis of LC behavior at bus bay stops. Therefore, employing evolutionary game models to observe bus departures from bus bay stops can provide valuable insights.

Furthermore, the analysis of existing literature has revealed the need for further investigation into various factors including action recognition of surrounding vehicles, multi-driving scenarios, communication delay, high-speed vehicle movement, uncertainty arising from external interference, driving scenarios involving intersection crossings, and

additional decision-making variables such as fuel economy.

7.4. EGT-based traffic control

To improve the accuracy of evolutionary game models, future research should focus on reducing assumptions and incorporating additional variables into the revenue function, considering factors such as efficiency and safety. Furthermore, studying the coordination of different groups within an extensive network of intersections is necessary. Additionally, addressing the modeling and solution of TSC in response to sudden, transient peak demand events, including those related to emergencies or disasters, is an important research gap that needs to be filled.

Various evolutionary game frameworks have been proposed in the literature, which have the potential to model the phenomena of behavioral differences among pedestrians, vehicles, and authorities in a group based on their behavior. Evolutionary game frameworks, however, are often integrated into microsimulation platforms with numerous calibration parameters. Consequently, evolutionary game frameworks become increasingly difficult to generalize as pedestrian and vehicle interaction models become more sophisticated. A modified game-theoretic model should be developed to resolve these limitations.

Though evolutionary game models have shown promising results, further studies are necessary to improve their performance. Most studies have only considered the social learning paradigm when modeling intersection crossing decisions using evolutionary game models. It may be possible to incorporate the individual learning paradigm into game theory as a follow-up study, which may prove advantageous to understanding pedestrians' crossing behavior. Furthermore, researchers should proceed cautiously when generalizing one's sample and results to others. A future study should examine other samples for additional evidence and convergence. In summary, the future evolutionary game model will require even more parameters to facilitate simulations than the models currently available in the literature. Therefore, future studies should investigate alternative methods for obtaining the probability of a motorist (or a pedestrian) choosing crossing over yielding, such as the rough set theory, which would reduce the number of parameters while preserving the model's behavior.

7.5. EGT-based congestion pricing

Considering travelers' heterogeneity is another crucial aspect when assessing dynamic congestion pricing schemes. Current models often use group-based variations, but future studies can explore user classes based on value of travel time and information acquisition rate. Additionally, further investigation is needed to examine different aspects of evolutionary game theory models in the context of heterogeneous users and elastic demand.

Implementation challenges persist in applying dynamic congestion pricing schemes to realistic transportation networks. Future research should explore issues such as non-existence and non-uniqueness of equilibrium solutions, stochastic dynamic route choice behavior, distance-based congestion pricing, and parallel computing for large-scale applications.

An addition to the aforementioned gaps in knowledge, there are several unanswered questions. These include understanding the behavior of the system with emergency trips, addressing privacy concerns, identifying all market segments, evaluating equity implications of entry rejections, handling re-entry requests from rejected travelers, and developing congestion pricing/priority mechanisms for various geographic regions and road types.

7.6. EGT-based traffic routing

As transport networks grow more complex, studying their dynamic behavior becomes crucial. Current research focuses on long-term

dynamic behavior within an evolutionary game framework, but considering varying system costs introduces a different game and may yield inaccurate solutions. Developing benchmarks for dynamic vehicle routing problems and incorporating stochastic factors into routing strategies are important areas for future research. Additionally, integrating objectives like profit maximization, risk minimization, fuel consumption reduction, and driver satisfaction into models should be explored. Furthermore, attention should be given to fleet management beyond routing to enhance service levels and overall performance.

In the context of Vehicular Ad-Hoc Networks (VANETs), evolutionary game model-based routing solutions have been proposed, but further work is needed. Developing a hybrid routing scheme combining relay-based, broadcast-based, and cluster-based routing is a promising research direction. However, hybrid routing requires addressing complexities and additional routing metrics. Additionally, trajectory prediction for routing strategies is an underexplored area due to limited information in highly dynamic vehicle networks.

7.7. EGT-based transport safety

In the past, drivers' multitasking behavior and traffic safety have been studied concerning utility maximization. The current studies make use of arbitrary collision parameters and utility functions for defining utility, but defining utility in the future using naturalistic driving behavior observed in shock waves during braking and stopping may be an interesting empirical challenge. In the current research, several studies have primarily focused on how vehicles interact with each other as leaders and followers. Future research may extend evolutionary game models to analyze how vehicles interact within platoons and how liability rules may impact rear-end crashes involving platoons.

Further, there is a lack of prediction regarding the severity of the expected collisions in most current traffic conflict models. To improve transport safety, it is necessary to fill this void because not only are vehicle collisions counted, but their severity must be taken into account. A bridge between these two gaps can be achieved in two ways. The first method estimates the conditional probability that a Collision outcome will occur in each severity category, whereas the second method simultaneously estimates both collision frequency and severity. These two approaches are highly dependent upon insights into collision occurrence, which unfortunately are not well understood/explored due to the complexity associated with road-user behavior. Accordingly, researchers can focus on conducting naturalistic driving studies, which enhance their understanding of how road users behave under normal and conflictual conditions and in collision situations.

From an EGT perspective, fewer articles provide speed recommendations based on collision/accident prevention. However, these studies considered single-lane roads without overtaking options. Further investigation is necessary to determine the optimal speed position on a multi-lane highway.

8. Conclusion

As an initial step, this paper gave background information for non-specialist readers, which includes a basic overview of GT and EGT. Evolutionary game theoretical contributions have been reviewed for traveler's choice behavior, traffic control and management, routing operation, and accidental safety in the transportation network, focusing on route choice, mode choice, lane selection, traffic signal control, congestion pricing, traffic routing, route guidance and drivers response, optimal driving strategy, optimal speed recommendations and possible collision prediction scenarios aligned with the emerging evolutionary game models. We conclude that evolutionary game models are most suitable for fulfilling behavioral and decision-making research directions in transportation systems. The main advantage of evolutionary models is the high scalability of a hierarchical proposal, hyperrationality, dynamicity, and easy adaptation to unexpected network behaviors

using previously learned strategies.

Declaration of Competing Interest

We don't have any kind of conflict of interest.

Data availability

Not Applicable.

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